

MCA Semester – IV
Research Project – Final Report

Name	SANKETH K S
Project	Predicting Customer Churn
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A study on “Predicting Customer Churn “

Research Project submitted to Jain Online (Deemed-to-be University)

In partial fulfillment of the requirements for the award of:

Master of Computer Application

Submitted by:

SANKETH K S

USN:

232VMTR00302

Under the guidance of:

Dr C S Jyothirmayee

(Faculty-JAIN Online)

Jain Online (Deemed-to-be University)

Bangalore

2024 - 25

DECLARATION

I, *SANKETH K S*, hereby declare that the Research Project Report titled “*Predicting Customer Churn*” has been prepared by me under the guidance of the *Dr C S Jyothirmayee*. I declare that this Project work is towards the partial fulfillment of the University Regulations for the award of the degree of Master of Computer Application by Jain University, Bengaluru. I have undergone a project for a period of Eight Weeks. I further declare that this Project is based on the original study undertaken by me and has not been submitted for the award of any degree/diploma from any other University / Institution.

Place: Madikeri

Date: 25 October 2025

SANKETH K S
USN: 232VMTR00302

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Abstract

A concise summary of model building:

In a rapidly evolving entertainment market, customer churn has emerged as a critical threat to subscription-based service providers. This study focuses on a Direct-to-Home (DTH) service provider that has been facing growing competition, resulting in declining customer retention rates. The company's primary goal is to predict customer churn—that is, the likelihood of a customer discontinuing their subscription—and to implement targeted, cost-effective retention strategies based on predictive insights.

A dataset comprising 11,260 customer accounts and 19 variables was analyzed to develop a predictive model. Several classification algorithms—Logistic Regression, Support Vector Machine (SVM), and Random Forest—were implemented to identify high-risk accounts. A comprehensive Exploratory Data Analysis (EDA) revealed key churn determinants, such as customer tenure, frequency of customer care interaction, service satisfaction scores, and preferred payment methods.

After rigorous model training and validation, Random Forest emerged as the optimal model, offering the best balance between predictive accuracy and interpretability. Business insights derived from the model informed actionable recommendations to reduce churn, including loyalty programs, targeted offers, and service quality improvements.

This project demonstrates how data-driven decision-making can enhance customer retention, reduce acquisition costs, and strengthen long-term profitability in the DTH industry.

Introduction

Introduction and Background:

The DTH sector has transformed home entertainment by providing digital television services directly to consumers via satellite. With increasing digital adoption, customers today have multiple options from various service providers. This high degree of competition means that customer loyalty has become a strategic differentiator.

For DTH companies, the cost of acquiring new customers—including installation, device subsidies, and marketing—is substantial. Therefore, retaining existing customers is vital for profitability. Retention is not only financially beneficial but also builds brand reputation and customer lifetime value. However, understanding why customers leave is a complex challenge that requires analyzing both behavioral and transactional data.

In this project, predictive analytics and machine learning are used to identify customers who are at risk of churn. By leveraging these insights, the company can design data-backed retention campaigns that are both effective and financially sustainable.

Problem Statement:

A Direct-to-Home (DTH) service provider is currently experiencing significant competition in the market, making it increasingly difficult to retain existing customers. Consequently, the company aims to create a model for predicting account churn and to offer tailored promotions to those at risk of leaving. The issue of account churn is particularly critical for this company, as a single account may represent multiple customers. Therefore, losing one account could result in the loss of several customers.

The objective of this project is to develop a churn prediction model for the company and to formulate business recommendations for a campaign designed to retain customers. The proposed campaign should be cost-effective and adhere to a predetermined budget allocated for customer retention efforts.

Objective of Study:

The objectives of the project are as follows:

➤ **Predictive Objective:**

Develop a machine learning model that accurately predicts customer churn based on demographic, behavioral, and transactional data.

➤ **Analytical Objective:**

Identify and interpret the key variables influencing customer churn to understand behavioral trends and customer dissatisfaction drivers.

➤ **Business Objective:**

Propose actionable and cost-effective marketing strategies to retain high-risk customers, thereby improving profitability and customer loyalty.

➤ **Strategic Objective:**

Enable the organization to adopt a proactive rather than reactive approach to customer management through predictive insights.

Literature Review

Company and industry overview:

The DTH industry operates within the broader broadcasting and entertainment ecosystem, delivering television content directly to customers' homes. Over the past decade, the industry has experienced rapid digitalization, an increase in high-definition (HD) and on-demand content, and intense competition from over-the-top (OTT) platforms like Netflix, Disney+, and Amazon Prime.

For traditional DTH companies, retaining existing customers is now as critical as acquiring new ones. The fixed operating costs (content rights, satellite bandwidth, licensing) mean profitability depends heavily on maintaining a stable subscriber base. Hence, churn prediction and retention optimization are now central to business strategy.

Overview of Theoretical Concepts:

Customer Churn:

Churn refers to the proportion of subscribers who stop using a service during a specific period. For subscription businesses, churn directly affects revenue growth and profitability.

Churn Prediction Models:

These are supervised machine learning classification models designed to predict the probability that a given customer will churn. Churn prediction involves using past customer behavior and service usage data to forecast future actions.

Key Modeling Techniques:

- **Logistic Regression:** Offers interpretability and works well as a baseline classifier.
- **Decision Tree:** Provides rule-based insights but may overfit.
- **Random Forest:** Uses multiple decision trees (ensemble method) to enhance accuracy and robustness.
- **Support Vector Machine (SVM):** Effective in handling high-dimensional spaces and non-linear separations.

Survey on the existing models:

Studies in telecom and subscription-based industries indicate that ensemble models, such as Random Forest and Gradient Boosting, often outperform linear models in predicting churn. However, logistic regression remains popular due to its interpretability.

In the DTH context, customer churn is influenced by factors such as service quality, technical issues, pricing, and customer support. Previous research has shown that a combination of behavioral (usage), transactional (payment, revenue), and demographic features provides the most predictive power.

Hence, this study applies a mix of interpretable (Logistic Regression) and high-performing (SVM and Random Forest) models to achieve both accuracy and actionable insights.

Data Report:

Variable	Description
AccountID	account unique identifier
Churn	account churn flag (Target)
Tenure	Tenure of account
City_Tier	Tier of primary customer's city
CC_Contacted_L12m	How many times all the customers of the account has contacted customer care in last 12months
Payment	Preferred Payment mode of the customers in the account
Gender	Gender of the primary customer of the account
Service_Score	Satisfaction score given by customers of the account on service provided by company
Account_user_count	Number of customers tagged with this account
account_segment	Account segmentation on the basis of spend
CC_Agent_Score	Satisfaction score given by customers of the account on customer care service provided by company
Marital_Status	Marital status of the primary customer of the account
rev_per_month	Monthly average revenue generated by account in last 12 months
Complain_l12m	Any complaints has been raised by account in last 12 months
rev_growth_yoy	revenue growth percentage of the account (last 12 months vs last 24 to 13 month)
coupon_used_l12m	How many times customers have used coupons to do the payment in last 12 months
Day_Since_CC_connect	Number of days since no customers in the account has contacted the customer care
cashback_l12m	Monthly average cashback generated by account in last 12 months
Login_device	Preferred login device of the customers in the account

Methodology

1. EDA and Business Implication:

Dataset Description & Data Collection Method:

➤ **Understanding how data was collected in terms of time, frequency and methodology:**

- Data has been collected for random 11,260 unique account ID, across gender and marital status.
- Looking at variables “CC_Contacted_L12m”, “rev_per_month”, “Complain_112m”, “rev_growth_yoy”, “coupon_used_112m”, “Day_Since_CC_connect” and “cashback_112m” we can conclude that the data has been collected for last 12 month.
- Data has 19 variables, 18 independent and 1 dependent or the target variable, which shows if customer churned or not.
- The data is the combination of services customers are using along with their payment option and also then basic individuals details as well.
- Data is mixed of categorical as well as continuous variables

➤ **Data Ingestion:**

Data set has 11,260 number of observations and 19 variables (18 independent and 1 dependent or target variable).

	AccountID	Churn	Tenure	City_Tier	CC_Contacted_LY	Payment	Gender	Service_Score	Account_user_count	account_segment	CC_Agent_Score	Marital_Status	rev_per_month	Complain_ly
0	20000	1	4	3.0	6.0	Debit Card	Female	3.0	3	Super	2.0	Single	9	1.0
1	20001	1	0	1.0	8.0	UPI	Male	3.0	4	Regular Plus	3.0	Single	7	1.0
2	20002	1	0	1.0	30.0	Debit Card	Male	2.0	4	Regular Plus	3.0	Single	6	1.0
3	20003	1	0	3.0	15.0	Debit Card	Male	2.0	4	Super	5.0	Single	8	0.0
4	20004	1	0	1.0	12.0	Credit Card	Male	2.0	3	Regular Plus	5.0	Single	3	0.0

Table1: Data-frame head with top 5 rows

➤ **Visual inspection of data (rows, columns, descriptive details):**

- Data has 11,260 rows and 19 variables.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11260 entries, 0 to 11259
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   AccountID                            11260 non-null  int64
1   Churn                                11260 non-null  int64
2   Tenure                               11158 non-null  object
3   City_Tier                            11148 non-null  float64
4   CC_Contacted_LY                      11158 non-null  float64
5   Payment                              11151 non-null  object
6   Gender                               11152 non-null  object
7   Service_Score                        11162 non-null  float64
8   Account_user_count                   11148 non-null  object
9   account_segment                      11163 non-null  object
10  CC_Agent_Score                       11144 non-null  float64
11  Marital_Status                       11048 non-null  object
12  rev_per_month                        11158 non-null  object
13  Complain_ly                          10903 non-null  float64
14  rev_growth_yoy                       11260 non-null  object
15  coupon_used_for_payment              11260 non-null  object
16  Day_Since_CC_connect                10903 non-null  object
17  cashback                             10789 non-null  object
18  Login_device                         11039 non-null  object
dtypes: float64(5), int64(2), object(12)
memory usage: 1.6+ MB
```

Table2: Dataset information

- Describing data: This shows description of variation in various statistical measurements across variables which denotes that each variable is unique and different.

	AccountID	Churn	City_Tier	CC_Contacted_LY	Service_Score	CC_Agent_Score	Complain_ly
count	11260.00000	11260.000000	11148.000000	11158.000000	11162.000000	11144.000000	10903.000000
mean	25629.50000	0.168384	1.653929	17.867091	2.902526	3.066493	0.285334
std	3250.62635	0.374223	0.915015	8.853269	0.725584	1.379772	0.451594
min	20000.00000	0.000000	1.000000	4.000000	0.000000	1.000000	0.000000
25%	22814.75000	0.000000	1.000000	11.000000	2.000000	2.000000	0.000000
50%	25629.50000	0.000000	1.000000	16.000000	3.000000	3.000000	0.000000
75%	28444.25000	0.000000	3.000000	23.000000	3.000000	4.000000	1.000000
max	31259.00000	1.000000	3.000000	132.000000	5.000000	5.000000	1.000000

Table3: Describing Dataset

- Showing Null Values in Dataset:

AccountID	0
Churn	0
Tenure	102
City_Tier	112
CC_Contacted_LY	102
Payment	109
Gender	108
Service_Score	98
Account_user_count	112
account_segment	97
CC_Agent_Score	116
Marital_Status	212
rev_per_month	102
Complain_ly	357
rev_growth_yoy	0
coupon_used_for_payment	0
Day_Since_CC_connect	357
cashback	471
Login_device	221

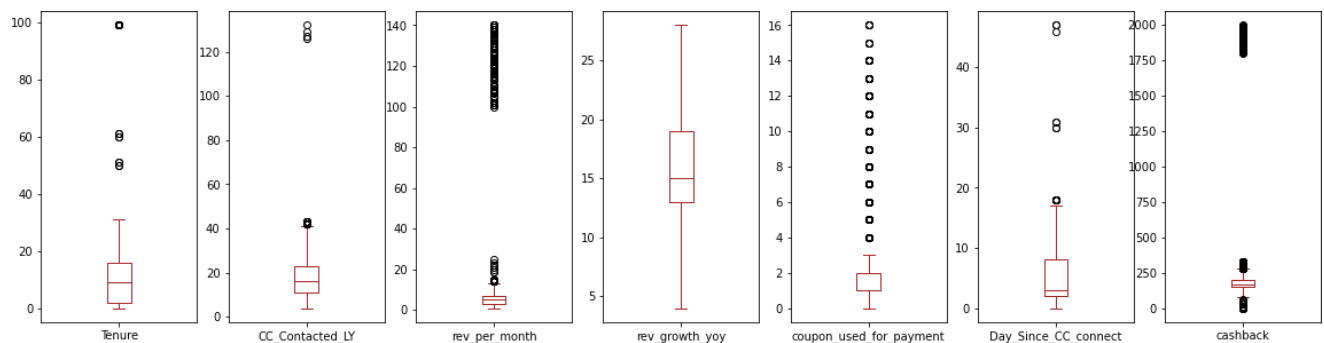
Table4: Number of null values in dataset

Exploratory Data Analysis:

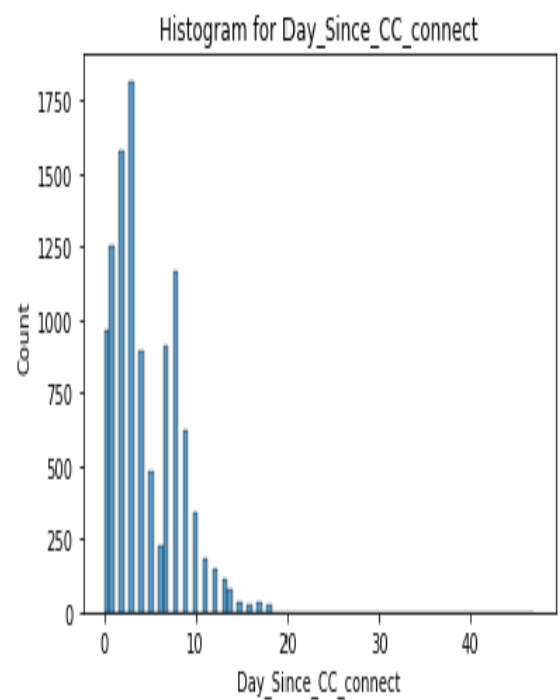
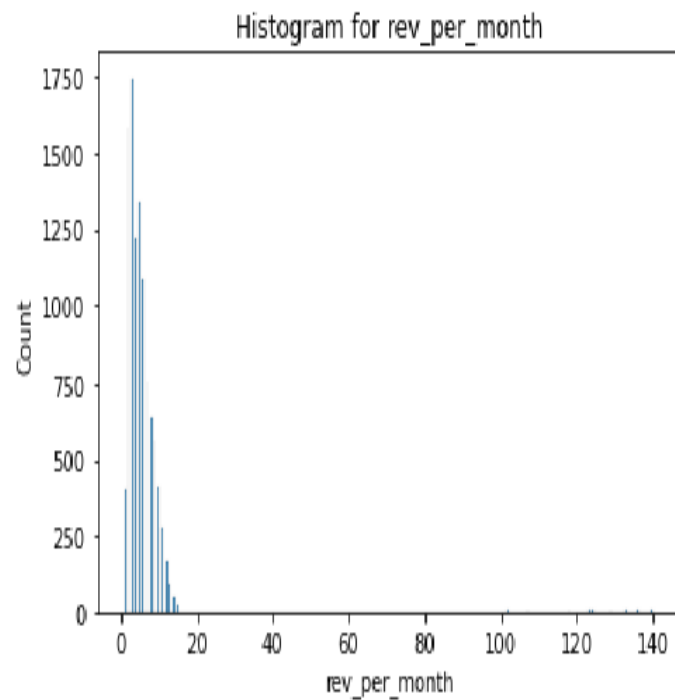
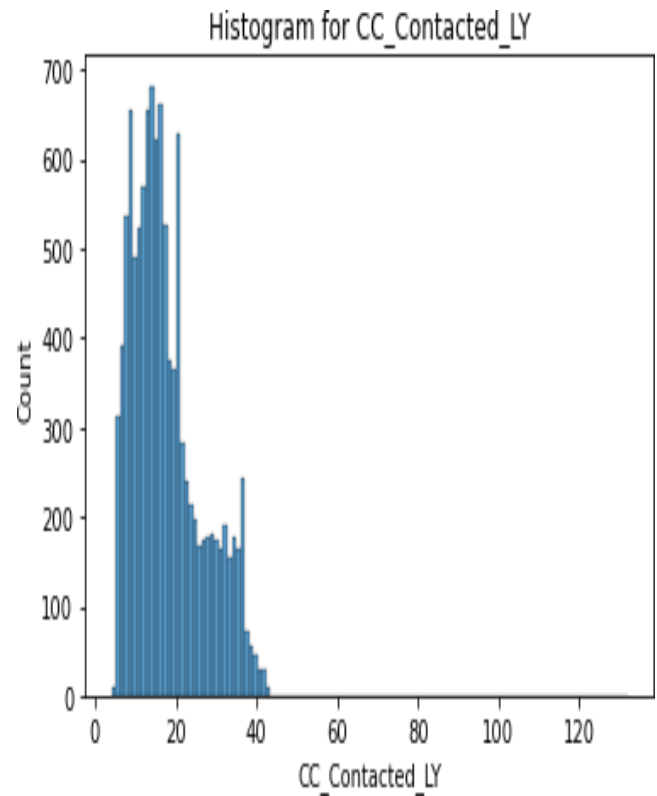
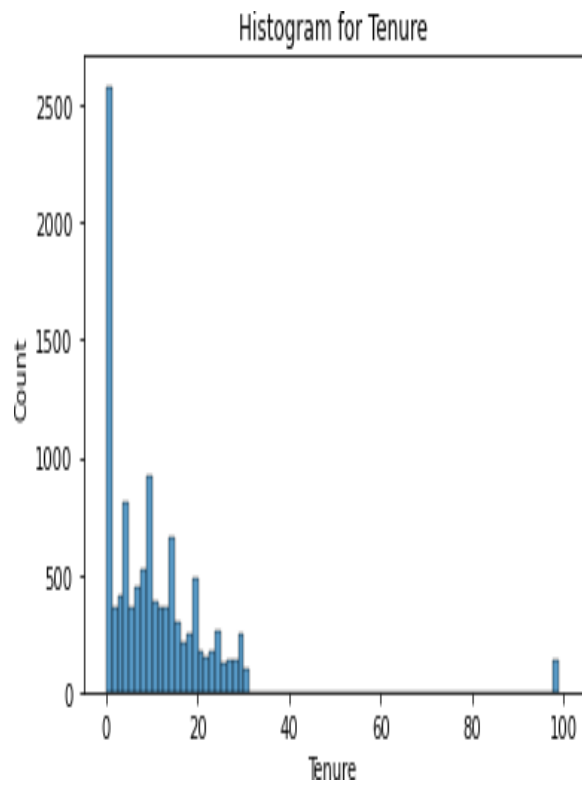
➤ Univariate analysis:

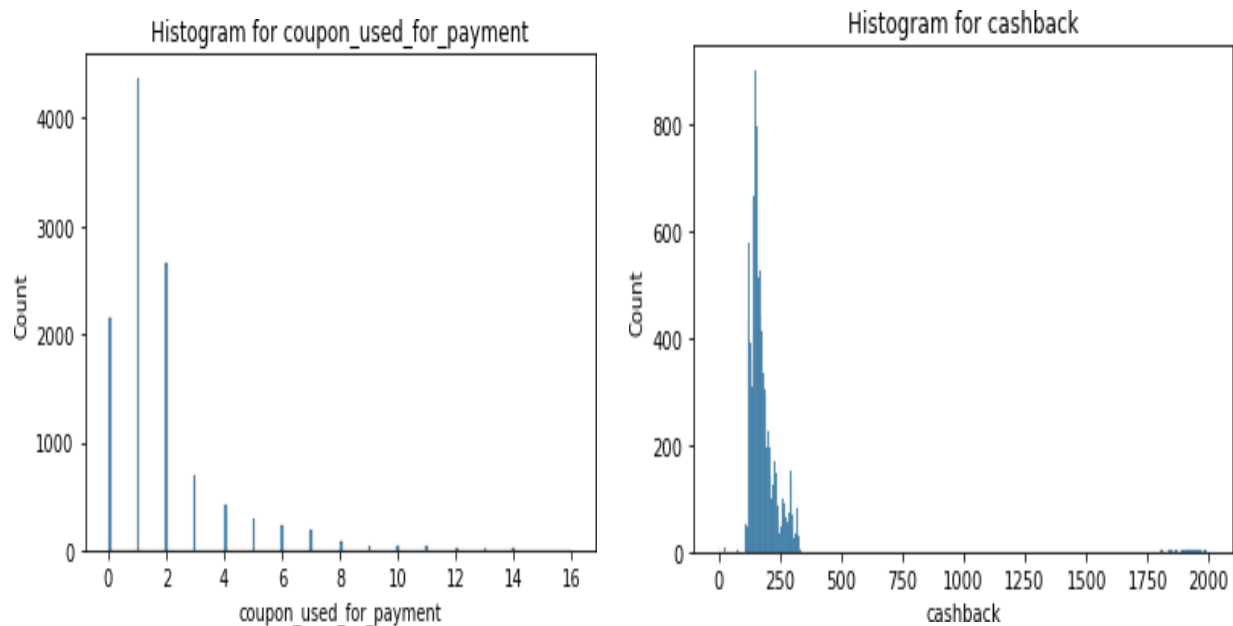
Univariate analysis is done for the purpose of observing distribution and spread for every continuous attribute and distribution of data in categories for categorical ones. It has been done by observing

- Box plots and histograms for continuous variables
- Count plots for categorical variables



Graph1: Box plot for numeric variable





Graph2: Continuous data -Histogram

	Skewness
Tenure	3.90
CC_Contacted_LY	1.42
rev_per_month	9.09
rev_growth_yoy	0.75
coupon_used_for_payment	2.58
Day_Since_CC_connect	1.27
cashback	8.77

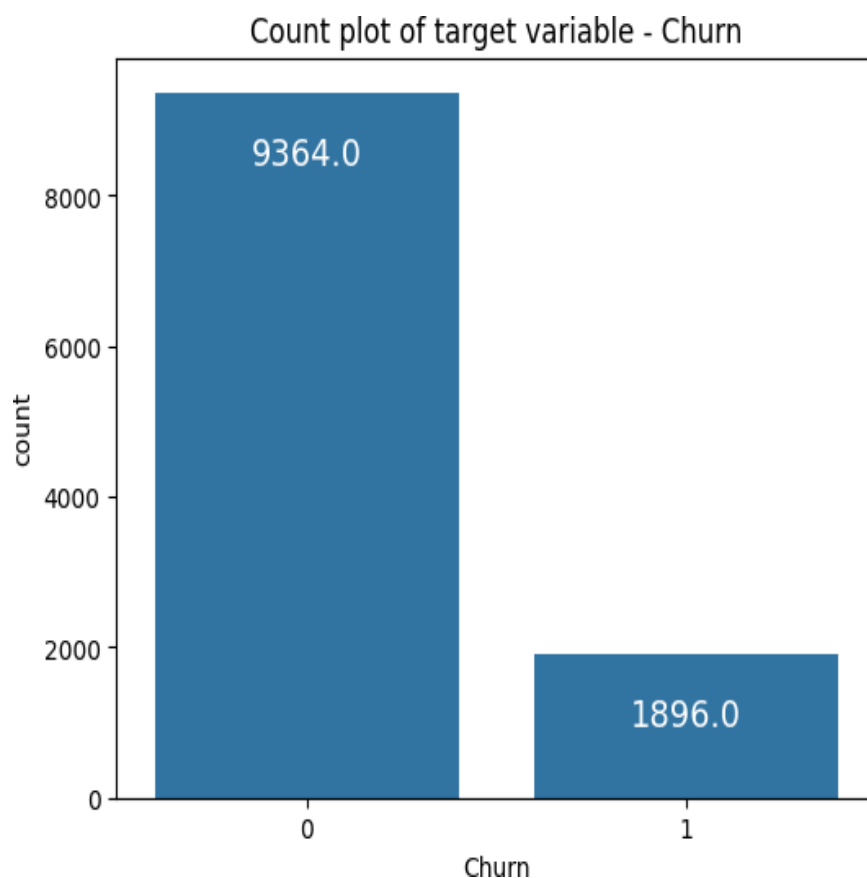
Table5: Skewness for numeric variables

Observations:

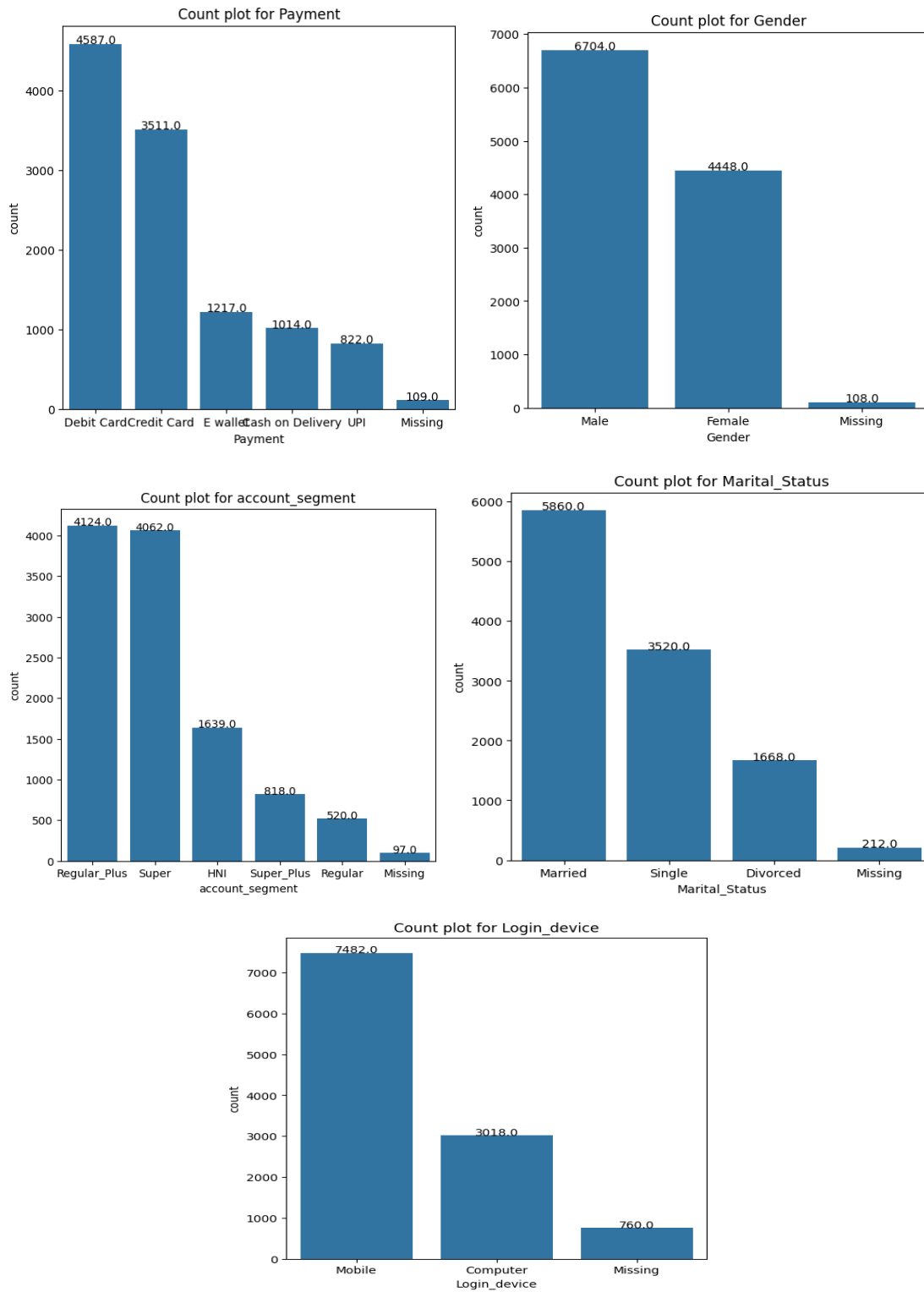
- All numeric variables, except for rev_growth_yoy, exhibit the presence of outliers. Some outliers for specific variables are situated near the whisker, while others are significantly distant from it, with no intermediate values. For example, rev_per_month shows a substantial gap between 30 and 100, indicating a lack of values within that range. The extreme outliers do not show a correlation

with the corresponding outliers in the cashback field. Although we cannot dismiss these outliers as erroneous, as they may pertain to hotels with a large number of rooms, models such as logistic regression are particularly sensitive to outliers and may perform poorly if these outliers are not addressed.

- Therefore, we will adopt two modeling approaches: one dataset will have outliers adjusted for models sensitive to outliers, while the other dataset will retain the outliers (left unchanged) for models that are resistant to outliers, such as Random Forest.
- The variable `coupon_used_for_payment` has a very narrow range of 0 to 16. Consequently, for this analysis, the outliers will remain unaddressed, similar to how categorical variables are treated.
- All numeric variables, with the exception of `rev_growth_yoy`, display a significant positive skew.



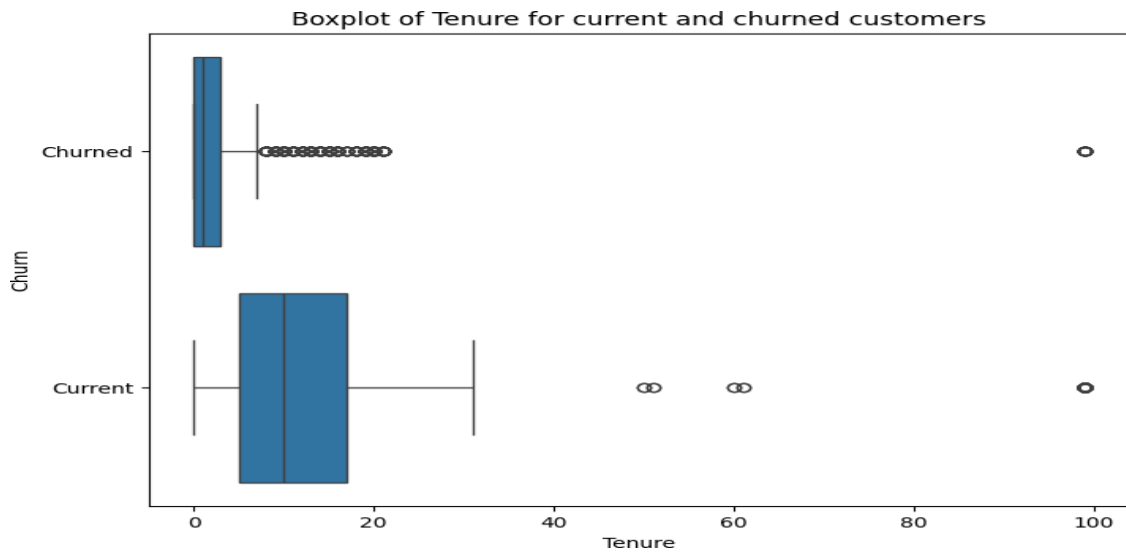
Graph3: Count plot of targeted variable 'Churn'



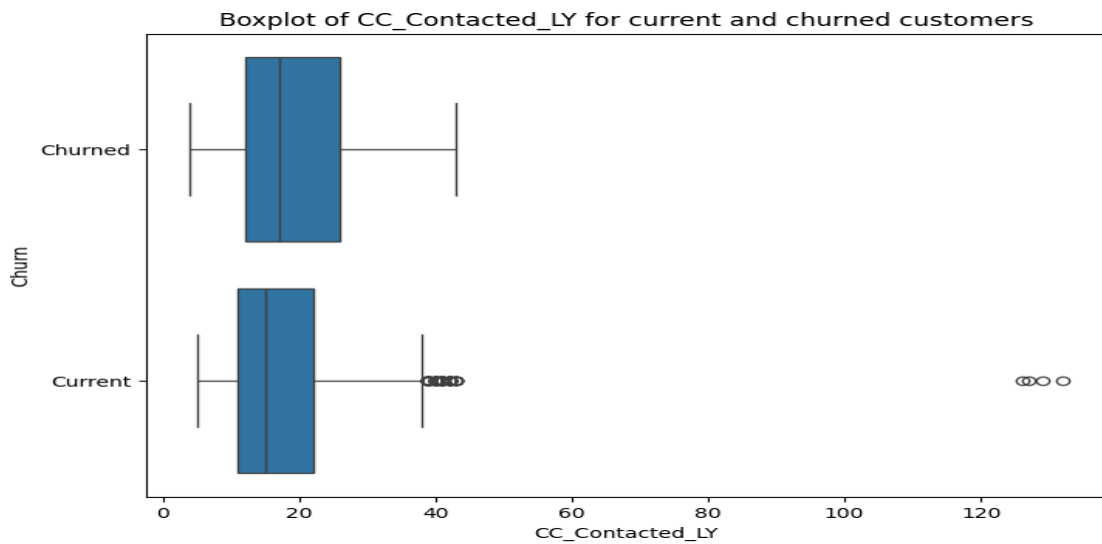
Graph4: Count plot for categorical variables

➤ **Bivariate Analysis:**

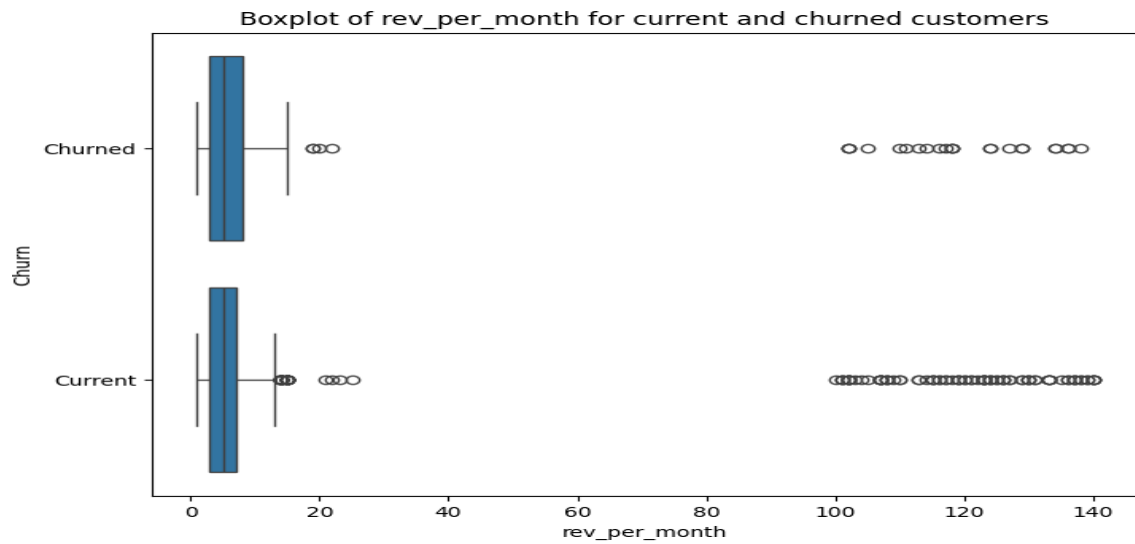
Bivariate analysis shows the relationship between two variables. Here, the predictor variables have been taken and their relationship with the target variable has been plotted. The influence of the predictor variables on the target variable can be observed in these bivariate plots.



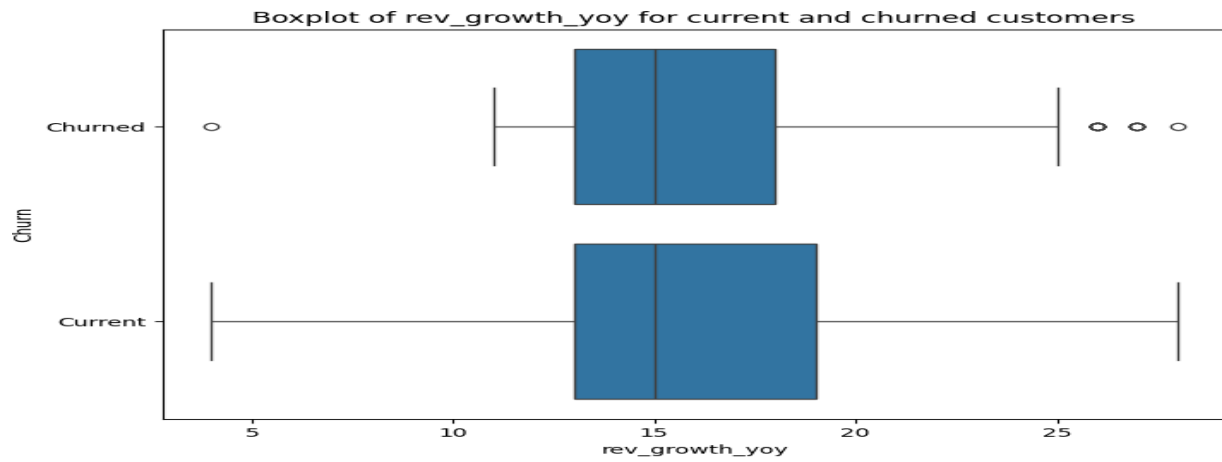
Graph5: Tenure Vs Churn



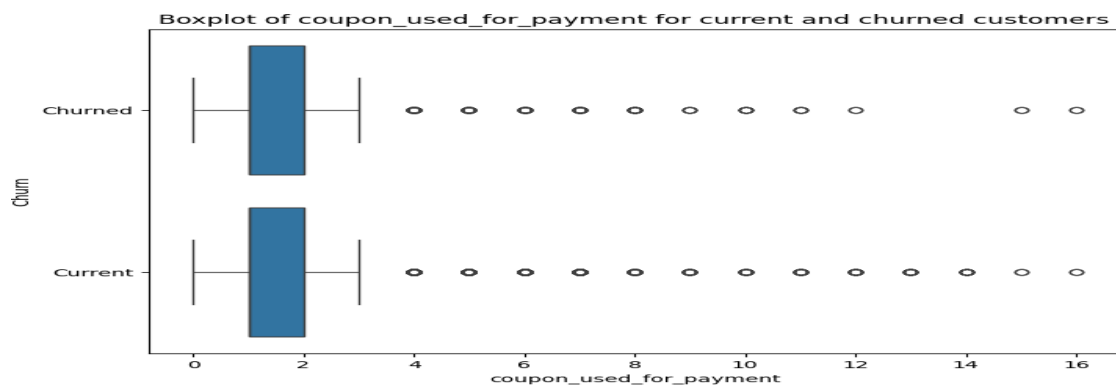
Graph6: CC_Contacted_LY Vs Churn



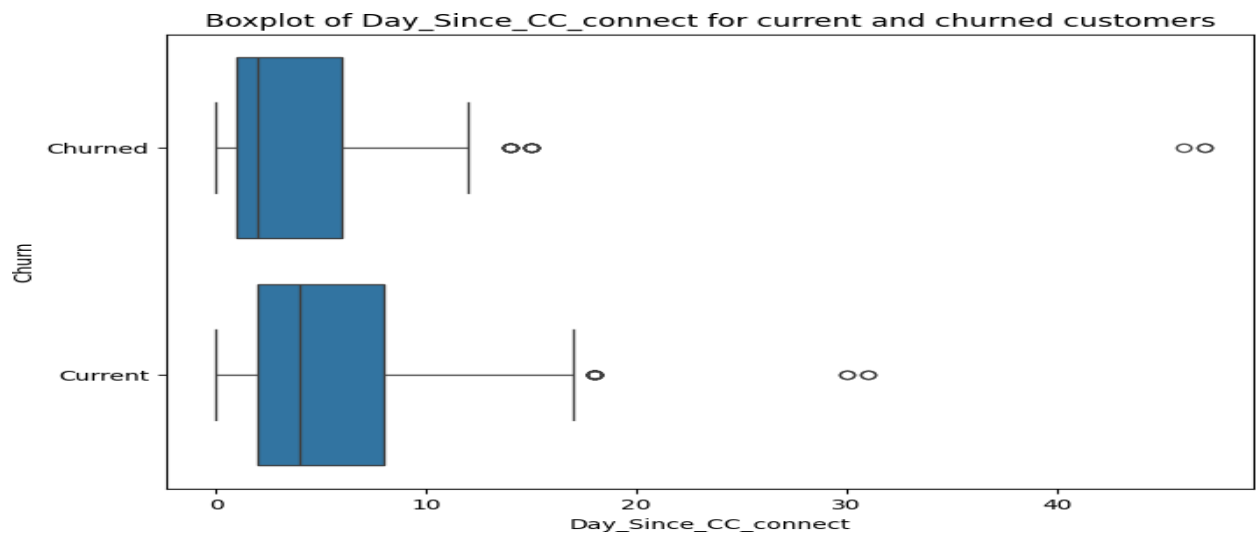
Graph7: rev_per_month Vs Churn



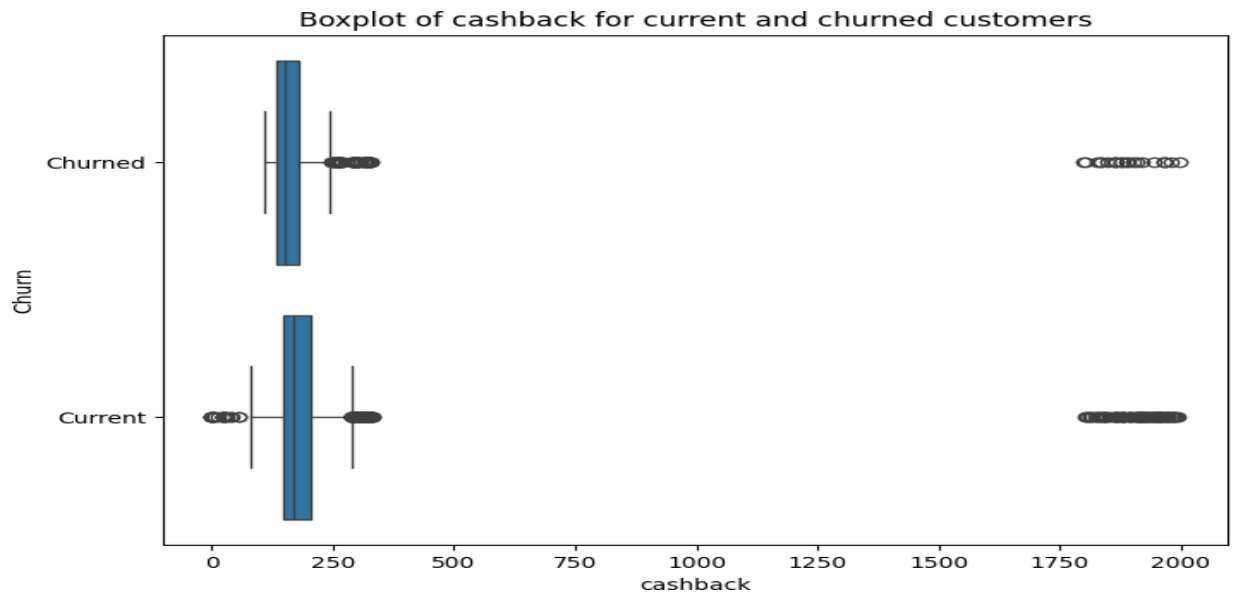
Graph8: rev_growth_yoy Vs Churn



Graph9: coupon_used_for_Payment Vs Churn



Graph10: Day_since_cc_connect Vs Churn

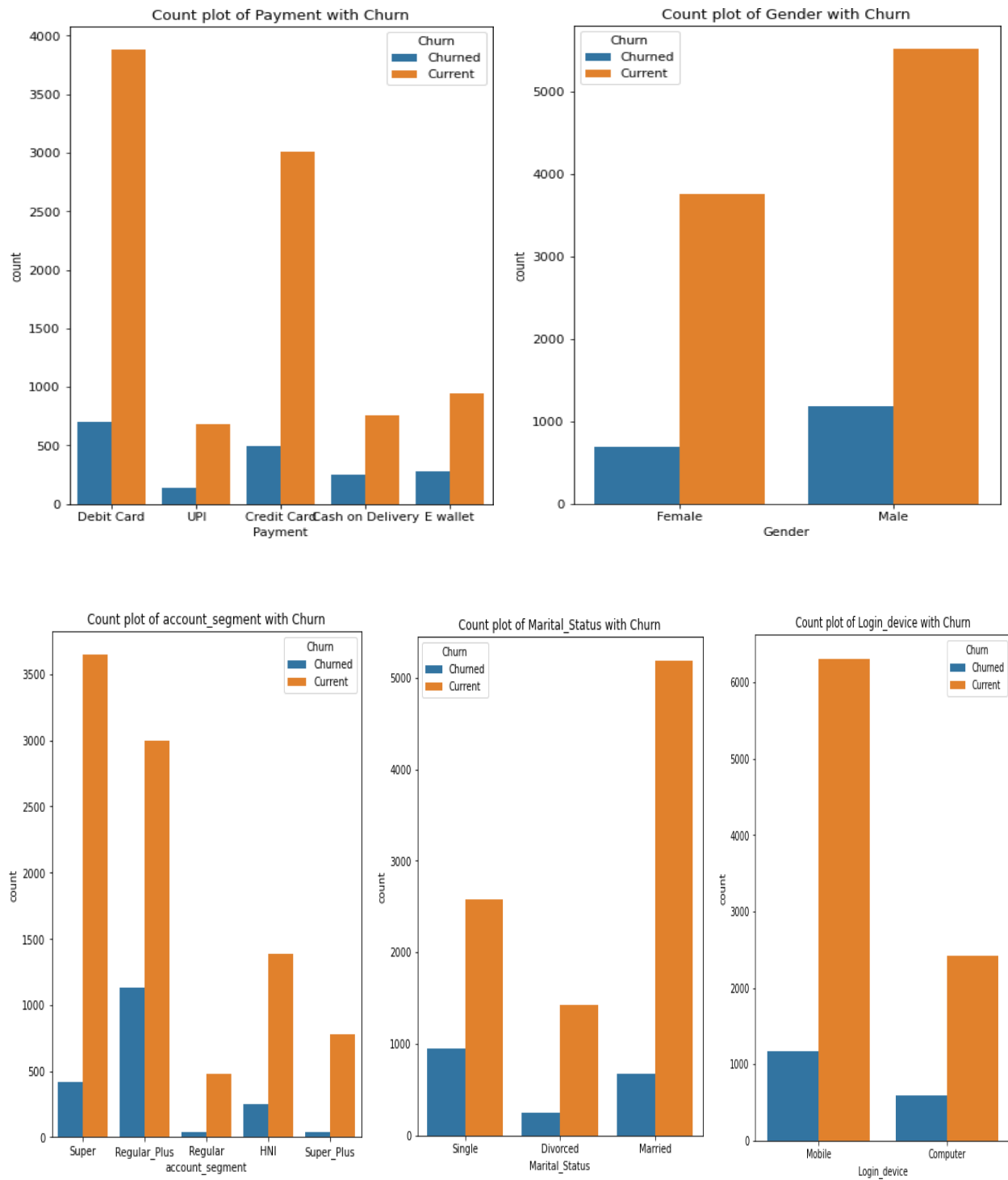


Graph11: cashback Vs Churn

Observations:

The plots above indicate that certain variables, including Tenure and Days_since_CC_connect, appear to have an impact on the target variable. When comparing the median lines for Churned and Non-churned observations against these variables, a noticeable difference is evident. In contrast, the boxplots for variables such as coupon_used_for_payment and rev_growth_yoy reveal minimal differences in the distributions between churned and non-churned groups.

Bivariate plots for categorical variables Vs churned:

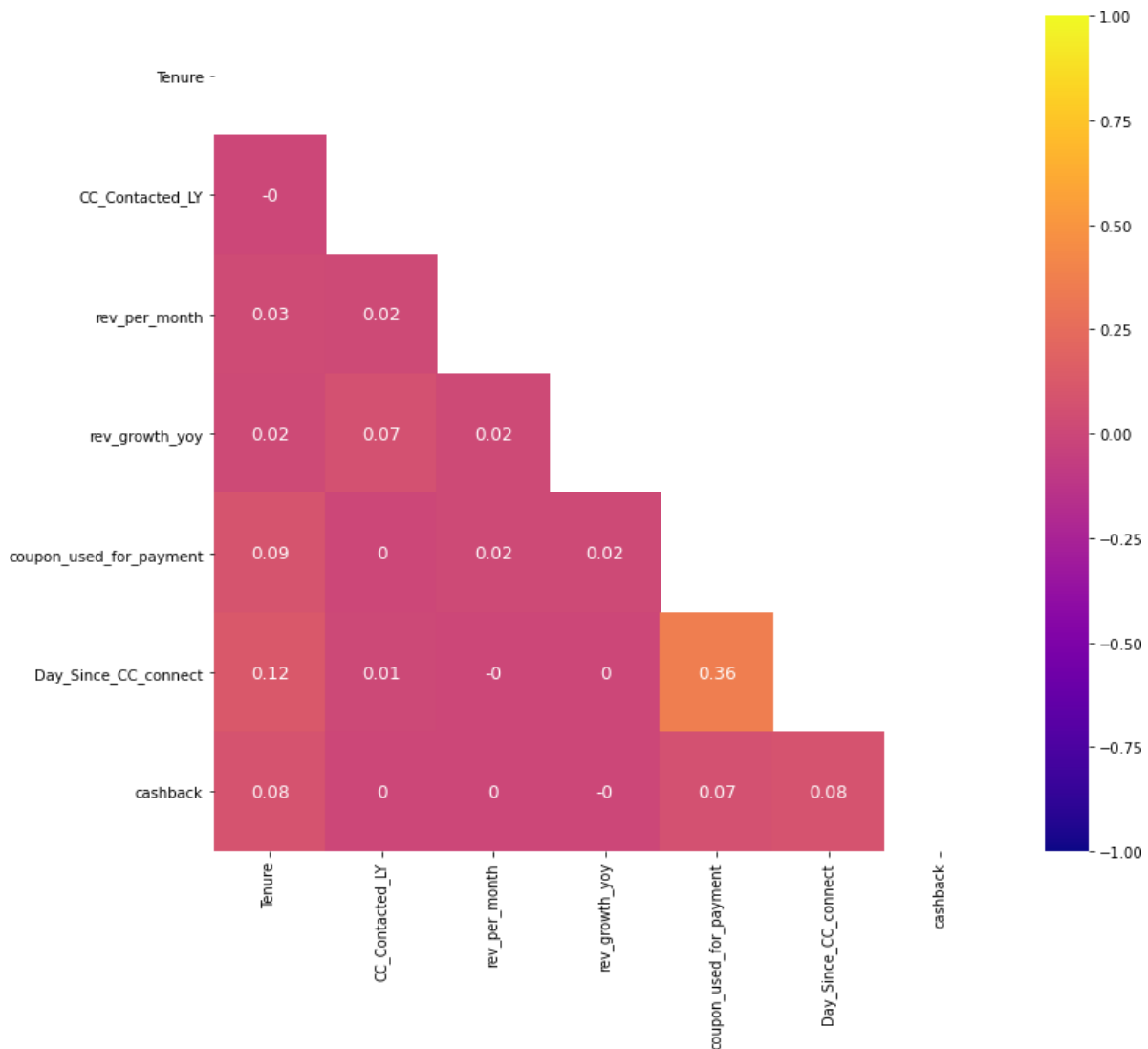


Graph12: Bar chart for categorical variables

➤ Multivariate analysis:

Correlation Matrix:

Heat map shows the correlation (pearson's) between various numeric predictors in the dataset. The purpose of doing a correlation matrix is to check if any of the variables have a strong correlation that can contribute to multi-collinearity. If there is a strong correlation > 0.80 between two variables, one of them can be dropped as it would not add much to the model's performance.



Graph13: Correlation heatmap

Observation: None of the numeric variables show a strong correlation.

Business Insights Derived from EDA:

- **Customer Feedback:** A significant 78% of customers have rated the service at 3 or below on a scale of 5. Similarly, 61% of customers have given customer care agents a score of 3 or less. This suggests a general trend of dissatisfaction or minimal satisfaction regarding both service quality and customer care interactions.
- **Tenure and Churn Correlation:** The bivariate histogram illustrating the relationship between Tenure and Churn reveals a high churn rate among customers with shorter tenures. Specifically, approximately 51.85% of customers with tenures between 0 and 1 have churned. It is essential to investigate the underlying reasons for this churn. As tenure increases, the histogram indicates a decline in the churn rate.
- **Account Segment and Churn Relationship:** Customers enrolled in the Regular_Plus plan appear to experience higher churn rates. A comparative analysis between this plan and similar offerings from competitors within the same pricing range could help determine if adjustments to the plan or pricing are necessary. Additionally, gathering and analyzing customer feedback from this segment may provide insights into the reasons for the elevated churn.
- **Monthly Revenue and Churn Correlation:** The churn rate among high-revenue customers is slightly higher than that of lower-revenue customers. This trend raises concerns for the DTH provider, as it indicates not only an increase in churn but also a loss of valuable high-revenue clientele.
- **Days Since Customer Care Contact and Churn:** Churned customers have a shorter duration since their last contact with customer care compared to active customers. This suggests that churn tends to occur shortly after customers reach out to customer care.
- **Customer Care Engagement Last Year and Churn:** Churned customers contacted customer care more frequently in the previous year than active customers.
- **Correlation between User Count and Churn** The percentage of accounts with user counts of 5 and 6 is higher among churned customers than among those who remain active.

- Correlation between Payment Method and Churn A greater proportion of churned customers have utilized E-wallets and Cash on Delivery as their payment methods compared to active customers.
- Correlation between City Tier and Churn Churned customers are more likely to reside in Tier 3 cities than their active counterparts. The DTH provider should investigate whether these cities experience greater competition than Tier 1 cities.
- Correlation between Marital Status and Churn Single customers exhibit a higher churn rate compared to those who are married or divorced.

2. Data Cleaning and Pre-Processing:

➤ Removal of unwanted variables:

- Any variable that possesses distinct values for each observation, such as the AccountID field, does not contribute to the model since it merely serves as an identifier for each observation.
- Additionally, any variable that remains unchanged across all or the majority of observations does not enhance predictive power. Analysis of the histogram (for numeric variables) and count plots/value counts (for categorical variables) indicates that there are no variables with a constant value across all observations.
- The variable with the highest percentage of missing values is cashback, which contains 4% nulls; therefore, no columns will be eliminated based on this criterion.
- Moreover, if a predictor variable exhibits a strong correlation with another predictor variable, one of them may be discarded. The correlation heat map indicates that there are no strong correlations present, so no variables will be removed on this basis.
- AccountID, rev_growth_yoy, and coupon_used_for_payment have been eliminated.

```
Index(['Churn', 'Tenure', 'City_Tier', 'CC_Contacted_LY', 'Service_Score',
      'Account_user_count', 'CC_Agent_Score', 'rev_per_month', 'Complain_ly',
      'Day_Since_CC_connect', 'cashback', 'Payment_Credit_CARD',
      'Payment_Debit_CARD', 'Payment_E_wallet', 'Payment_UPI', 'Gender_Male',
      'account_segment_Regular', 'account_segment_Regular_Plus',
      'account_segment_Super', 'account_segment_Super_Plus',
      'Marital_Status_Married', 'Marital_Status_Single',
      'Login_device_Mobile'],
      dtype='object')
```

Table6: After removal of variable

➤ Missing Value treatment:

The treatment of missing values was conducted through KNN imputation, which is a distance-based approach. The following steps were necessary prerequisites for implementing KNN for missing value treatment:

- All variables must be numeric. Categorical variables of object type require appropriate encoding, either through label encoding or one-hot encoding. The columns that were one-hot encoded included nominal categorical variables. To prevent multicollinearity, the first encoded variable was excluded. The one-hot encoded variables included 'Payment', 'Gender', 'account_segment', 'Marital_Status', and 'Login_device'.
- Following the encoding process, the column names were modified to ensure they were concise, consistent, and free of blank spaces that arose from certain category values in the one-hot encoded columns (for instance, 'Payment_Credit Card').

```
Tenure      0
City_Tier   0
CC_Contacted_LY  0
Service_Score  0
User_Count  0
CC_Score    0
Rev_Permonth  0
Complain_LY  0
Days_Since_CC  0
Cashback    0
Payment_Creditcard  0
Payment_Debitcard  0
Payment_Ewallet  0
Payment_UPI  0
Gender_Male  0
ACSegment_Regular  0
ACSegment_Regularplus  0
ACSegment_Super  0
ACSegment_Superplus  0
Maritalstatus_Married  0
Maritalstatus_Single  0
Logindevice_Mobile  0
dtype: int64
```

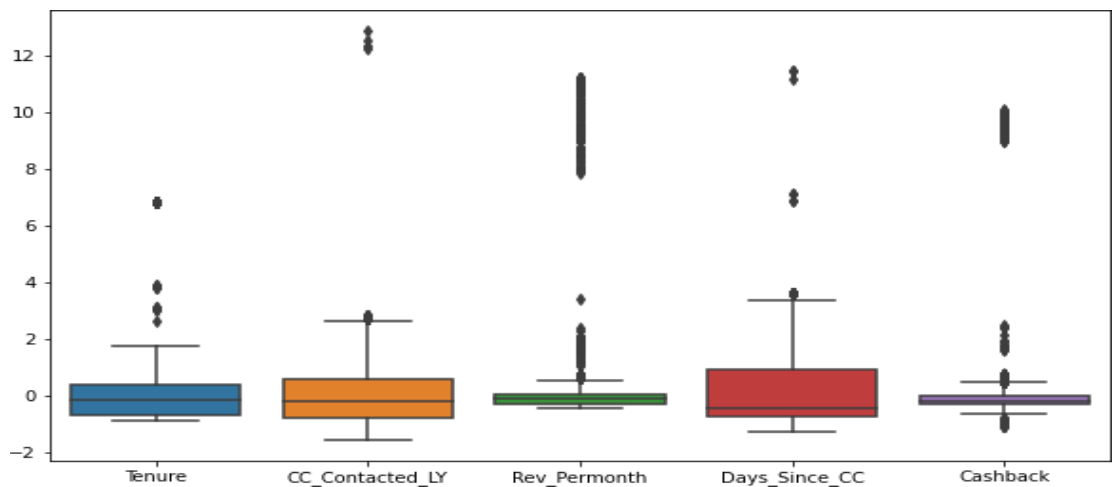
Table7: Null variables after KNN imputing

➤ **Variable transformation:**

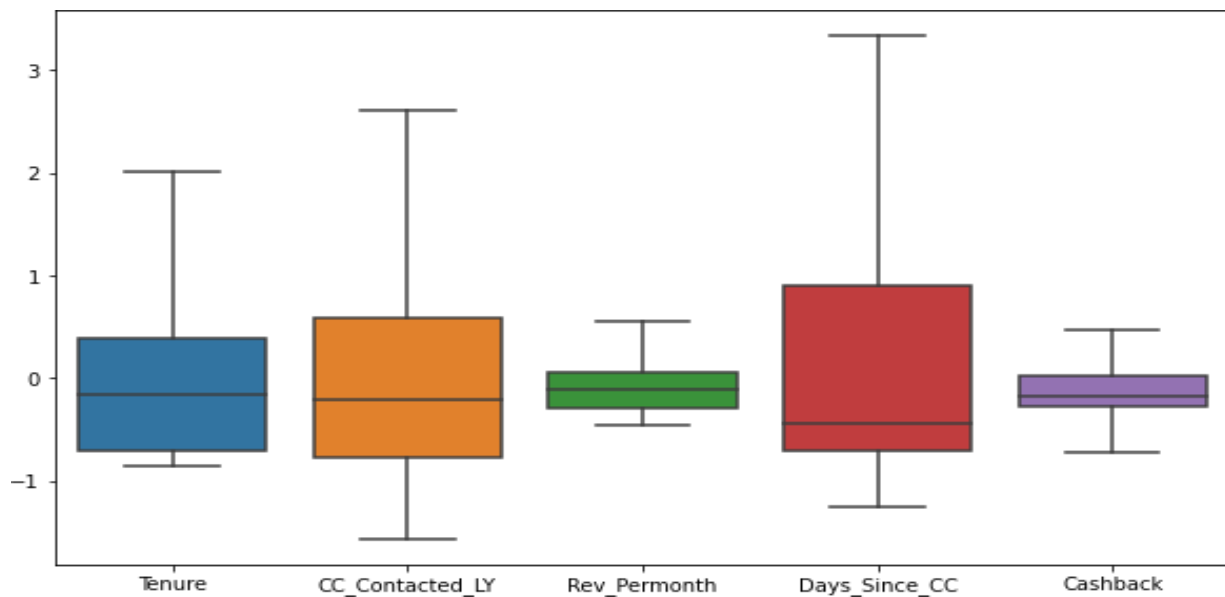
- The variables Payment, Gender, Account_Segment, Marital_Status, and Login_device are all categorical in nature. To facilitate analysis, these variables must be transformed into numeric format. Since the categories within these variables lack a specific order, one-hot encoding was employed. This process involves dividing the variable into several columns, with each column representing a binary value for each category present in the original attribute. Furthermore, to prevent multi-collinearity—which can lead to reduced model performance and challenges in interpretation—one of the columns resulting from the one-hot encoding is removed.
- The dataset has undergone scaling, which is essential for any distance-based algorithms such as KNN imputer. The scaled dataset is compatible with all models, regardless of whether they require scaled input.
- Currently, no additional transformations are anticipated for modeling purposes.

➤ **Outlier Treatment:**

The dataset includes both continuous and categorical variables. Since categorical variables represent different customer types, it doesn't make sense to apply outlier treatment to them. Therefore, outlier treatment will only be applied to the continuous variables.



Graph14: Before outlier treatment



Graph15: After outlier treatment

3. Model Building and Validation:

According to both visual and non-visual analyses, it seems we are facing a classification challenge in which the target variable must be classified as either "Yes" or "No." As data analysts, we have access to various algorithms that can be utilized to create a model aimed at predicting customer churn. These algorithms consist of:

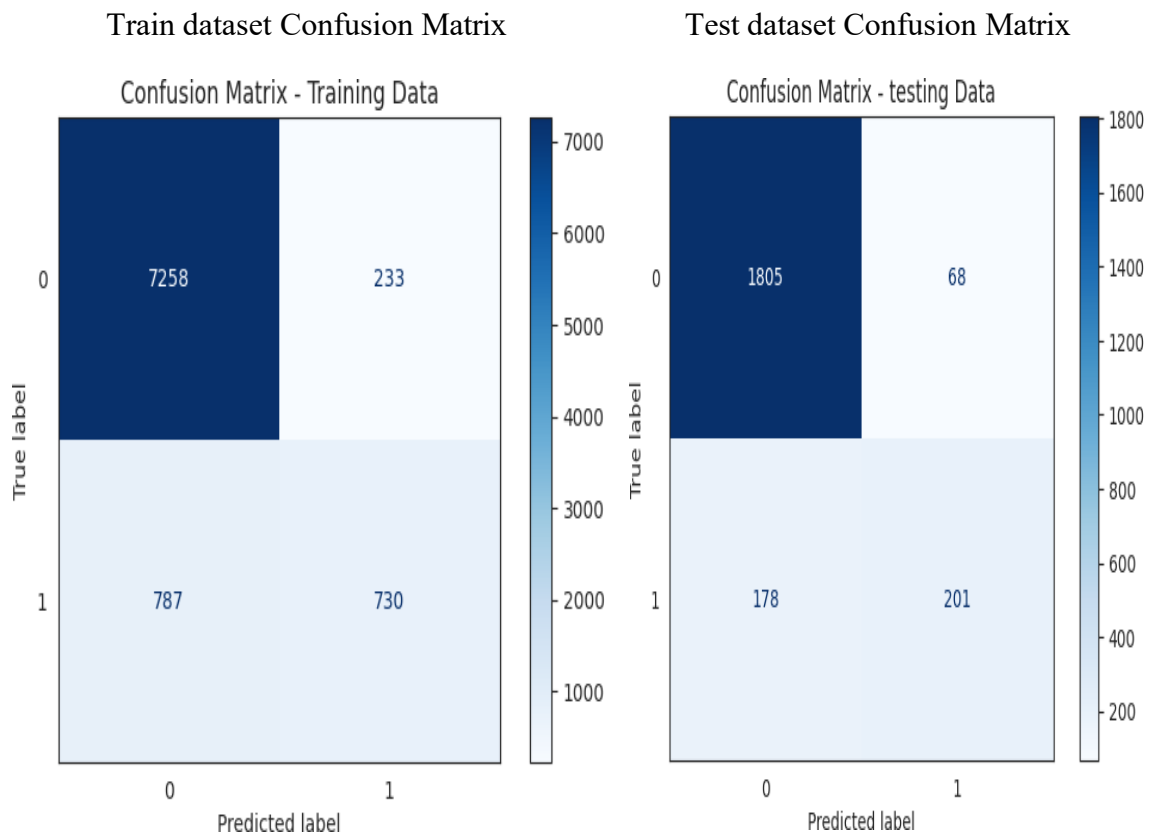
- **Logistic Regression:** Logistic Regression is one of the "white box" algorithms that helps determine probability values and corresponding cuts. Logistic regression is used to solve problems that give a corresponding probability output, allowing the corresponding point of the output of the target class to be determined. This model is used because it does not require assumptions and will classify faster.
- **Decision Tree:** A decision tree is a type of monitored machine learning used to classify or predict predictions based on how a previous set of questions were answered. Models are a form of surveillance learning. In other words, the model is trained and tested with a set of data containing the desired classification. This model is selected as one of the models because of the white box algorithm. Even in the data set, there are no zero values in the data record. You can make the formed tree visible.

- **Random Forest:** Random Forest is a monitored algorithm for machine learning and is often used in classification and regression problems. It constructs decisions, creates trees with different samples, adjusts their majority for classification, and adjusts the mean of regression. Grid search is used to understand the best parameters for the model. This model selection is due to the efficiency and higher accuracy of processing large data records.
- **Support Vector Machine:** Support Vector Machine is an algorithm popular for machine learning and can be used for classification and regression. If there is a higher dimension, it works well. It's very versatile as there are various core functions that you can specify to work well on the specified.

➤ **Logistic Regression Model:**

LR_model1 is the best model out of the Logistic regression models:

The detailed information regarding base model, best performing model, their train- test metrics, model interpretability is provided below.

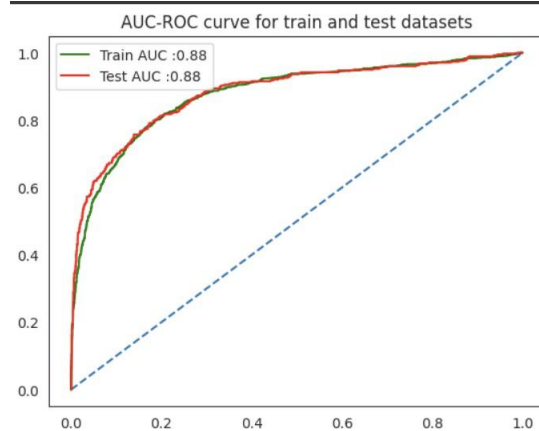


Train dataset Classification report

Test dataset Classification report

Model Performance: Logistic Regression Accuracy: 0.8868					Model Performance: Logistic Regression Accuracy: 0.8908				
Classification Reports:					Classification Reports:				
Logistic Regression:	precision	recall	f1-score	support	Logistic Regression:	precision	recall	f1-score	support
0	0.90	0.97	0.93	7491	0	0.91	0.96	0.94	1873
1	0.76	0.48	0.59	1517	1	0.75	0.53	0.62	379
accuracy			0.89	9008	accuracy			0.89	2252
macro avg	0.83	0.73	0.76	9008	macro avg	0.83	0.75	0.78	2252
weighted avg	0.88	0.89	0.88	9008	weighted avg	0.88	0.89	0.88	2252

Table8: Logistic regression-Base model and best model performance

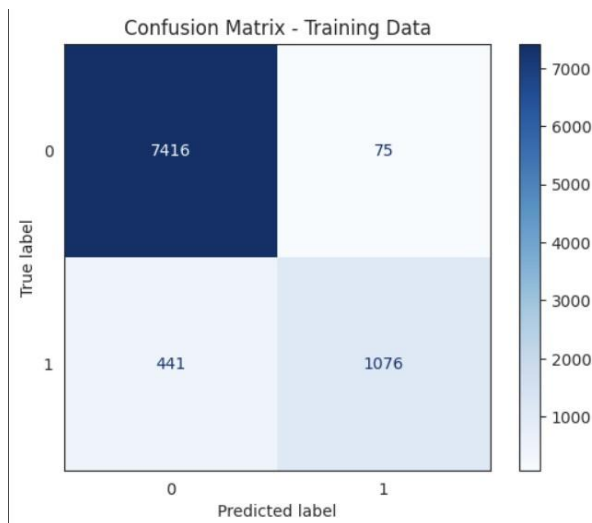


Graph 16: ROC Curve of LR model train set (L) test set (R)

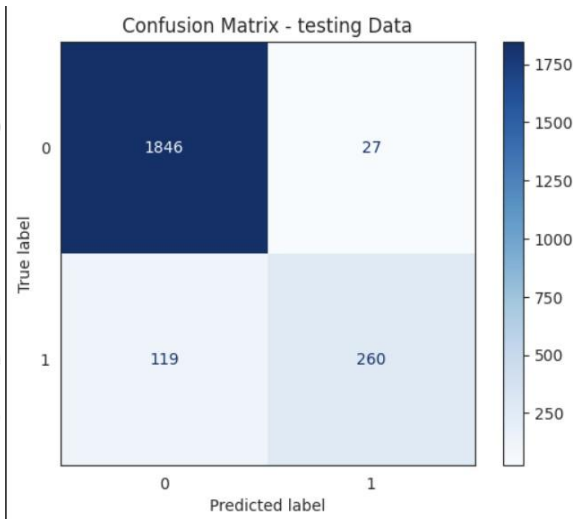
➤ Support vector machine (SVM):

SVM is a popular machine learning algorithm that can be used for classification as well as regression. It works well when there are higher dimensions as well. It is very versatile as there are different kernel functions that can be specified to work well with the given data.

Train dataset Confusion Matrix



Test dataset Confusion Matrix



Dataset Classification report

```

Model Performance:
SVM Accuracy: 0.8317

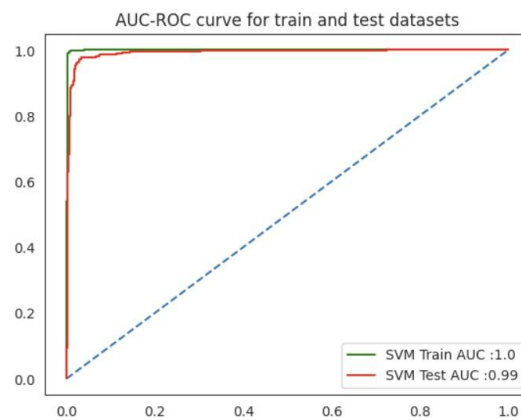
Classification Reports:

SVM:

```

	precision	recall	f1-score	support
0	0.83	1.00	0.91	1873
1	0.00	0.00	0.00	379
accuracy			0.83	2252
macro avg	0.42	0.50	0.45	2252
weighted avg	0.69	0.83	0.76	2252

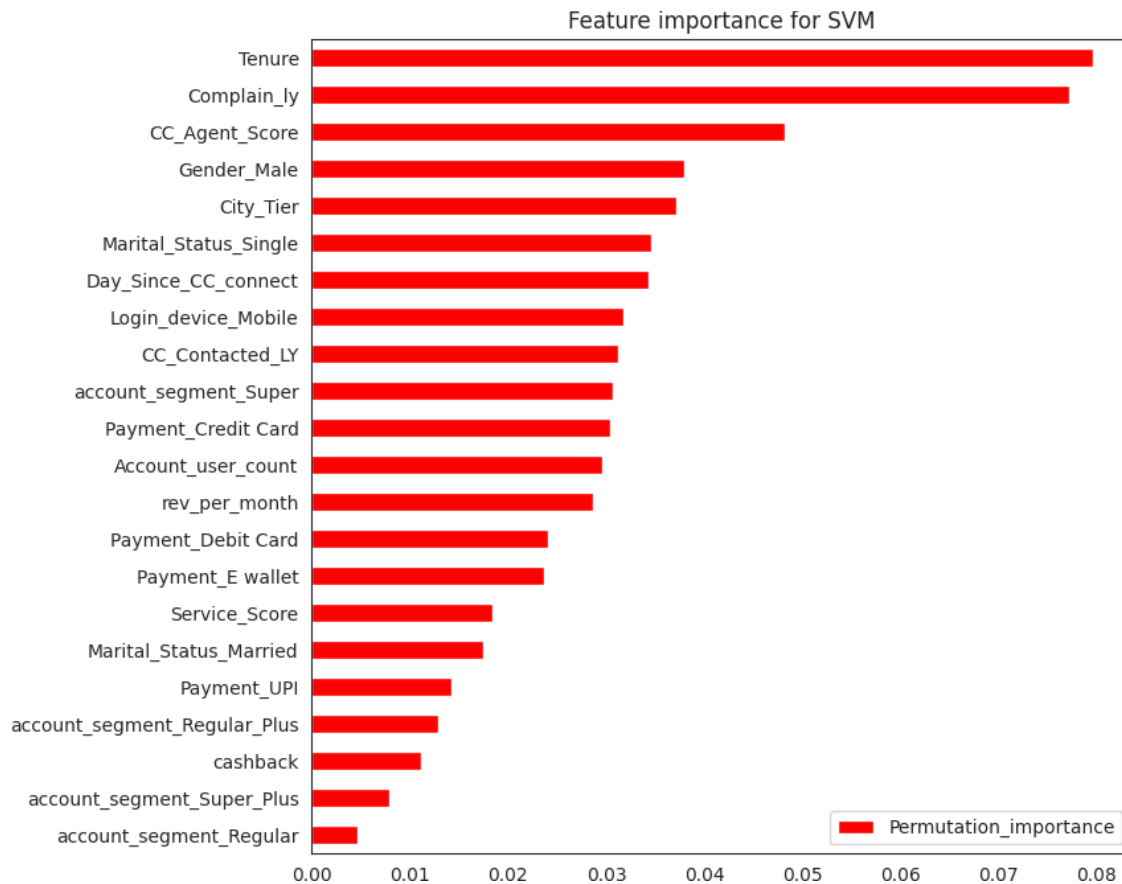
Table 9: SVM model and best model performance



Graph 17: ROC Curve of SVM model train set (L) test set (R)

Model interpretation:

SVM has given a reasonably good performance on train and test datasets. There has been no overfitting or underfitting of the model as the train and test dataset performances have been comparable. SVM's model coefficients attribute can be used only if linear kernel is used.



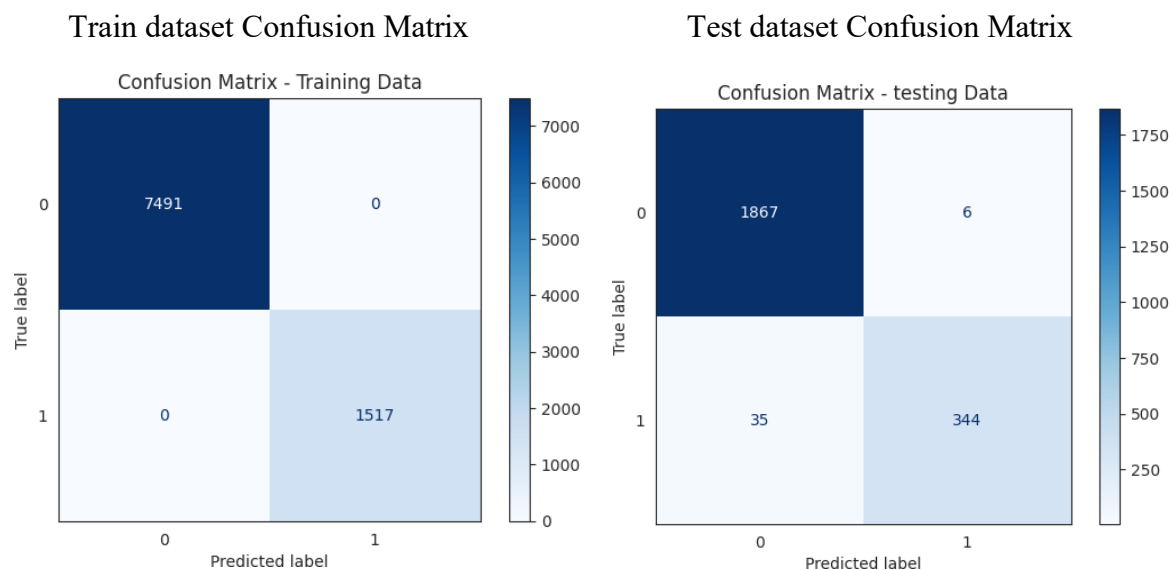
Graph 18: Feature importance from the best SVM model

In contrast to Logistic Regression and Linear Discriminant Analysis, this model has picked Tenure to be the most important feature followed by Complaints last year and Customer care score. From EDA, we can observe that for lower tenures especially within the first year, the churn is higher. Hence, once a customer has been acquired, the first year is very important to keep the customer satisfied. The next risk indicator of customer churn per this model is if a complaint has been registered in the previous year. From EDA, if a complaint was registered, the risk of churn is high.

The customer satisfaction score is also important per the SVM model. This is consistent with the patterns and insights that emerged out of EDA.

➤ Random Forest:

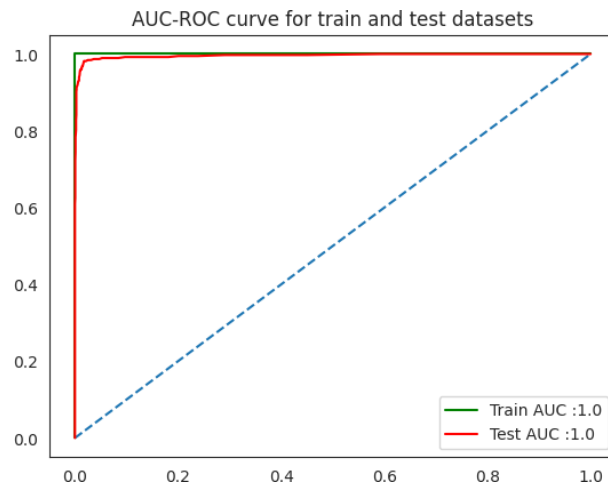
Random forest is an ensemble machine learning algorithm that uses bootstrapping to reduce variance in the underlying decision trees. It also selects only a subset of features for each node split decision. Since it is an ensemble of trees with varying features for each node, each tree is different from another. Random forest is resistant to outliers and does not require the data to be scaled.



Dataset Classification report

Model Performance:				
Random Forest Accuracy: 0.9729				
Classification Reports:				
Random Forest:				
	precision	recall	f1-score	support
0	0.98	0.99	0.98	1873
1	0.96	0.88	0.92	379
accuracy			0.97	2252
macro avg	0.97	0.94	0.95	2252
weighted avg	0.97	0.97	0.97	2252

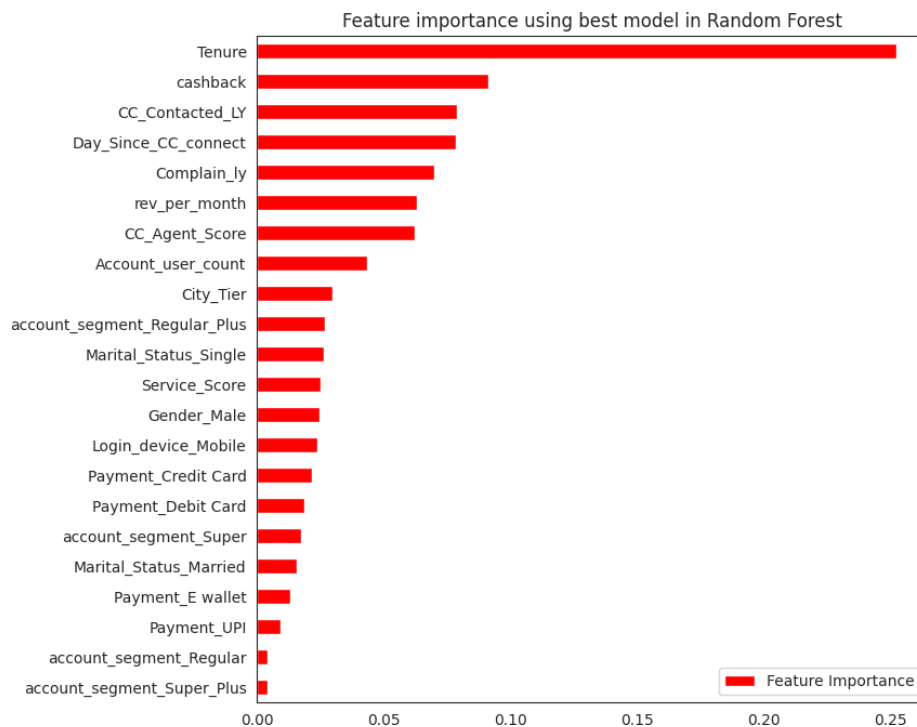
Table 10: Random forest model and best model performance



Graph 19: ROC Curve of Random forest model train set (L) test set (R)

Model interpretation:

This model has overfitted both in the default base model as well as the hyper parameter tuned model. The tuned model in fact performed worser than the base model. Hence in the final comparison, this cannot get selected as best model.



Graph 20: Feature importance from the best Random Forest model

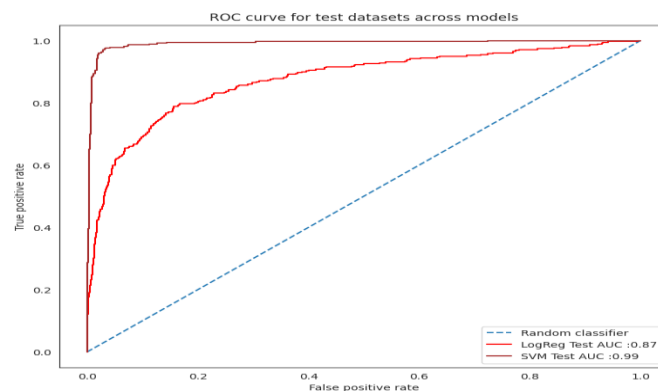
Similar to SVM, this model has also picked Tenure to be the most important feature. However, the next two parameters in terms of importance are not the same amongst, SVM and Random Forest. Since the performance for this model is not good the feature importance may not be reflective of actual data.

This model has picked Tenure to be the most important feature followed by Customer care contacted previous year and Days since customer care last contact.

- From EDA, we can observe that for lower tenures especially within the first year, the churn is higher. Hence, once a customer has been acquired, the first year is very important to keep the customer satisfied.
- The next important parameter to predict customer churn per this model is number of times customer care was contacted by the customer. Per EDA, the median and third quantile of number of times customer care was contacted previous year is higher for churned customers compared to active/current customers.
- Churned customers had contacted customer care more recently before churning than active customers. Per EDA, median days since last customer connect is higher for active customers. Churned customers had contacted customer care recently before churning

Interpretation of the most optimum model and its implication on the business:

The most optimum model is Random Forest model. This is evident because of the performance metrics obtained from the model used as shown above and the ROC curve combined as shown below.



Graph 21: Comparison of ROC Curve of all model train set (L) and test set (R)

Modeling Algorithms Used:

Model	Description	Reason for Use
Logistic Regression	Simple, interpretable classifier	Benchmark model
Support Vector Machine (SVM)	Handles complex decision boundaries	High-dimensional capability
Random Forest	Ensemble model combining multiple trees	High accuracy, robust to noise/outliers

Performance Comparison:

Model	Interpretability	Overfitting	Performance	Remarks
Logistic Regression	High	Low	Moderate	Baseline model; clear interpretability
SVM	Moderate	Low	Good	Stable and consistent
Random Forest	Low	Slight	Best	High accuracy and AUC

Key Predictors Identified:

Rank	Feature	Business Interpretation
1	Tenure	Shorter tenure → higher churn
2	Customer Care Contact (L12M)	Frequent complaints → dissatisfaction
3	Service Score	Low service rating → churn likelihood
4	Payment Method	COD and E-wallet users → inconsistent renewals
5	Account Segment	Regular_Plus → value dissatisfaction

ROC Curve and Accuracy Comparison:

- **Random Forest** achieved the best ROC-AUC and overall predictive performance.
- **SVM** demonstrated balanced train-test accuracy without overfitting.
- **Logistic Regression** served as a reliable baseline but less accurate.

The **Random Forest** model achieved the highest **ROC-AUC score**, with good generalization between train and test sets, confirming minimal overfitting.

Results and discussion

This section presents the key findings obtained from exploratory data analysis, machine learning model evaluation, and general insights gathered through observation and interpretation. The results not only highlight technical performance but also provide actionable knowledge that can drive customer retention strategies in the telecom industry.

Findings Based on Observations:

The initial exploratory data analysis (EDA) helped uncover critical behavioral patterns and customer characteristics linked to churn. These findings are significant because they offer a business-contextual understanding of who is more likely to leave and why:

- Early-tenure customers (≤ 1 year) are most vulnerable to churn.
- Low service satisfaction directly correlates with customer attrition.
- Churned customers had higher customer care engagement frequency.

These findings provide actionable insights into target segments for intervention, allowing the business to craft personalized campaigns and value propositions for high risk groups.

Findings Based on Data Analysis:

In addition to observational trends, statistical and machine learning analyses revealed deeper patterns in the dataset and confirmed the hypotheses derived from EDA.

- Churners exhibit reactive behavior, contacting support just before leaving.
- Payment inconsistency and Tier 3 locality influence churn.
- Married customers are more stable subscribers, possibly due to family-oriented usage.

General Findings:

From both the exploratory and predictive stages of analysis, several general conclusions can be drawn that apply beyond the dataset and align with industry-wide patterns.

- The Random Forest model effectively captures non-linear patterns in customer behavior.
- Tenure, Service Quality, and Payment Type are dominant churn predictors.
- There is significant potential for churn reduction through personalized retention strategies and service experience enhancement.

Insights:

- Business must increase the tenure of their customer which can be done by initiating some loyalty programs or special pricings for loyal customers
- Right targeting of the customers in Tier 2 and 3 cities
- The best possible time is to give some sort of promos at $T = 2$ (2 units of tenure)
- More focus on customers who use CC, COD and E wallet mode of payment, since this is a very critical segment
- Mostly customer rated "3" for the services provided by the business
- Mostly customer rated "3" for the interactions they have customer care representatives.
- Transaction via UPI and e-wallet is very low.
- Maximum churn is from the account segment "Regular+"
- Customers with marital status is "single" contributes max towards churn
- Any complaints raised in last 12 months doesn't show any impact toward churn.

- Tenure and cashback are directly proportional to each other.
- Computer usage is more in tier 1 city followed by tier 3 and tier 2 city

Conclusion and Recommendations

Recommendation based on findings:

Based on the data analysis, model performance, and business insights derived throughout this project, several targeted recommendations are proposed to help telecom companies proactively address customer churn.

- **Loyalty Programs:**

Implement tenure-based reward points or exclusive pricing to encourage long-term retention.

- **Targeted Marketing:**

Focus promotional offers on customers in Tier 2 and Tier 3 cities, particularly those using E-wallet and COD payments.

- **Customer Experience Management:**

Establish a priority customer support team for high-revenue accounts and Track post-service satisfaction using automated feedback forms.

- **Payment Optimization:**

Promote digital and auto-renewal options (UPI, standing instructions) to minimize unintentional churn.

- **Customer Engagement:**

Personalized thank-you messages, holiday vouchers, and referral discounts can create emotional loyalty.

Suggestions for areas of improvement:

While the project outcomes are robust, there are several areas where the data quality, model robustness, and operational deployment can be improved.

- **Incorporate Real-Time Data:** The current model is built on historical/static data. Integrating real-time usage patterns, complaint logs, or web/app behavior can improve prediction timeliness and accuracy.
- **User Feedback Integration:** Including textual data such as customer reviews, complaint transcripts, or NPS scores through natural language processing (NLP) could provide a richer understanding of customer sentiment.
- **Model Explainability:** While XGBoost is accurate, tools like SHAP (SHapley Additive Explanations) could be implemented to offer interpretable explanations to business users and customer-facing teams.
- Improve CRM integration to monitor real-time churn risk using predictive alerts.
- Collect more behavioral data such as viewing patterns and complaint topics.
- Enhance data granularity by including temporal data (monthly revenue trends).

Scope for future research:

This project opens up several potential directions for further study and enterprise application.

- **Time Series-Based Churn Modeling:** Churn behavior may evolve over time. Future models can explore temporal data, tracking how behavior changes across billing cycles or contract periods.
- **Customer Segmentation + Churn Modeling:** Combining clustering techniques with classification models can yield segment-specific churn insights, allowing for differentiated marketing strategies.

- **Predicting Lifetime Value (CLV):** Future work can extend the churn model to estimate customer lifetime value, allowing prioritization of high-risk, high-value customers for retention efforts.
- **Causal Inference in Retention Campaigns:** Research can move from predictive to prescriptive analytics, testing which retention interventions causally reduce churn (e.g., via A/B testing or uplift modeling).

Conclusion:

This capstone project successfully demonstrates that data-driven churn prediction is a powerful strategic tool for the DTH industry. The study successfully developed a machine learning-based framework to predict customer churn in the telecom industry. Starting from data exploration and cleaning to model building and evaluation, each phase was guided by both technical rigor and business relevance.

The Random Forest model, supported by strong EDA insights, effectively identifies at-risk customers. By acting on the predictive signals—such as low satisfaction scores and short tenure—the company can reduce churn, optimize marketing budgets, and enhance overall customer lifetime value.

Proactive retention, backed by predictive modelling, ensures that the DTH provider not only sustains its existing customer base but also strengthens its competitive positioning in a challenging market.

Python Code Snippet

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib
import matplotlib.pyplot as plt
```

```
CC = pd.read_excel("/content/LoyaltyVision Analytics (1).xlsx")
CC.head()
```

```
[ ] ## Shape of dataset
CC.shape
```

```
(11260, 19)
```

```
[ ] CC.size
```

```
213940
```

```
[ ] ## High level information
CC.info()
```

```
[ ] CC.duplicated().sum()
```

```
[ ] CC.isnull().sum()
```

```
[ ] ## Proportion of nulls in predictors out of total size of the predictor dataset expressed as percentage
CC_predictor = CC.drop(['AccountID','Churn'],axis=1)
```

```
round(((CC_predictor.isnull().sum().sum()/ CC_predictor.size) * 100 ),2)
```

```
np.float64(1.4)
```

```
[ ] ## Nulls by rows
```

```
CC_predictor.isnull().sum(axis = 1)
```

```
CC_predictor_temp = CC_predictor[CC_predictor.isnull().sum(axis = 1) >= 3]
CC_predictor_temp
```

```
[ ] CC_predictor_temp = CC_predictor[CC_predictor.isnull().sum(axis = 1) == 2]
CC_predictor_temp.shape
```

Data Cleaning

```
## Tenure field
```

```
CC.Tenure.value_counts()
```

```
##Replacing # in tenure with nulls. There are 116 rows where Tenure has the value '#'. Filling this with np.nan  
CC['Tenure'] = CC['Tenure'].replace('#',np.nan)
```

```
CC.Tenure.value_counts(dropna=False)
```

```
## Checking for City_tier column
```

```
CC.City_Tier.value_counts(dropna=False)
```

```
## Checking for payment
```

```
CC.Payment.value_counts(dropna=False)
```

```
CC['Gender'] = CC['Gender'].replace('M','Male')
```

```
CC['Gender'] = CC['Gender'].replace('F','Female')
```

```
CC.Gender.value_counts(dropna=False)
```

```
CC['account_segment'] = CC['account_segment'].replace( {'Regular +': 'Regular_Plus', 'Super +': 'Super_Plus', 'Super Plus': 'Super_Plus', 'Regular Plus': 'Regular_Plus'} )
```

```
CC.account_segment.value_counts(dropna=False)
```

After data cleaning null check

```
CC_predictor = CC.drop(['AccountID','Churn'],axis=1)

print("\nTotal number of nulls in all predictor fields ")
print(CC_predictor.isnull().sum().sum())

print("\nTotal % nulls in all predictor fields ")
print(round(((CC_predictor.isnull().sum().sum()/ CC_predictor.size) * 100 ),2))

## Percentage nulls by column in predictor dataset
print("\n\nProportion of nulls by column - multiply by 100 for percentage")
print(CC_predictor.isnull().sum().sort_values(ascending = False)/CC_predictor.index.size)

## Rows that have more than 4 null fields
print("\n\nRows that have more than 4 null fields")
CC_predictor_temp = CC_predictor[CC_predictor.isnull().sum(axis = 1) >= 4]
CC_predictor_temp
```

Univariate Analysis for continuous data

Box plot for continuous variables

```
] plt.figure(figsize=(12,8))
   CC_predictor[num_cols].boxplot()
   plt.xticks(rotation=90)
   plt.title("Box plot for all numeric continuous variables");
```

```
## Making a separate list for numeric and categorical columns

num_cols = ['Tenure', 'CC_Contacted_LY', 'rev_per_month', 'rev_growth_yoy', 'coupon_used_for_payment', 'Day_Since_CC_connect', 'cashback']
cat_cols = ['City_Tier', 'Payment', 'Gender', 'Service_Score', 'Account_user_count', 'account_segment', 'CC_Agent_Score', 'Marital_Status', 'Complain_ly', 'Login_device']

## Get upper bound and lower bound
Q1 = CC_predictor[num_cols].quantile(0.25)
Q3 = CC_predictor[num_cols].quantile(0.75)
IQR = Q3 - Q1
UL = Q3 + 1.5*IQR
LL = Q1 - 1.5*IQR

print ("Proportion of outliers in numeric continuous columns \n")
(((CC_predictor[num_cols] > UL) | (CC_predictor[num_cols] < LL)).sum())/ CC_predictor.shape[0]
```

```
plt.figure(figsize=(12,8))
sns.boxplot(data=CC, y='cashback')
plt.xticks(rotation=90)
plt.title("Box plot for 'Cashback' variable");
```

```
plt.figure(figsize=(12,8))
CC_predictor[num_cols].drop('cashback',axis=1).boxplot()
plt.xticks(rotation=60)
plt.title("Box plot for numeric continuous variables except 'Cashback'");
```

```
CC[num_cols].plot(kind = 'box', subplots = True,
                  figsize = (20,5), color = 'brown');
#plt.title("Boxplot for numeric columns");
```

```
CC['coupon_used_for_payment'].value_counts()
```

Count plot for categorical variables

```
[ ] ## Count plot of target column

plt.figure(figsize=(6,5))
ax = sns.countplot(x='Churn', data=CC)

for p in ax.patches:
    ax.annotate(f'\n{p.get_height()}', (p.get_x()+0.4, p.get_height()), ha='center', va='top', color='white', size=14)
plt.title("Count plot of target variable - Churn");
```

Histogram for continuous variables

```
## Using histplot
for i in num_cols:
    plt.figure(figsize=(6,4))
    sns.histplot(data=CC, x=i)
    plot_title = 'Histogram for ' + i
    plt.title(plot_title)
    plt.show();
```

▼ Bivariate analysis - Numerical fields vs Churn

Histogram and boxplots with hue

```
[ ] for i in num_cols:
    sns.displot(x=CC[i], hue=CC['Churn'], kind='hist', kde=True, aspect=2)
    plot_title = "Histogram for " + i + " for Churned customers and Non-churned customers"
    plt.title(plot_title)
    plt.show();
```

```
for i in num_cols:
    plt.figure(figsize=(8,6))
    sns.boxplot(x=CC[i], y=CC['Churn'])
    plot_title = 'Boxplot of ' + i + " for current and churned customers"
    plt.title(plot_title)
    plt.show();
```

▼ Bivariate Analysis - Categorical vs Churn

Count plots and stacked bar

```
[ ] ## Countplot with hue

for i in cat_cols:
    plt.figure(figsize=(6,8))
    sns.countplot(data = CC, x=i, hue='Churn')
    plot_title = "Count plot of "+i+" with Churn"
    plt.title(plot_title)
    plt.show();
```

Distribution of coupon_used_for_payment by churn

```
plt.figure(figsize=(10,6))
sns.countplot(data = CC_temp, x='coupon_used_for_payment', hue='Churn')
plot_title = "Count plot of "+"Coupon used for payment"+" with Churn"
plt.title(plot_title)
plt.show();
```

Correlation Matrix

```
CC_num = CC[num_cols]
corr = CC_num.corr()
mask = np.triu(np.ones_like(corr, dtype = np.bool))

plt.figure(figsize = (12,11))
sns.heatmap(round(corr,2), annot=True, mask=mask, cmap='plasma', vmin=-1, vmax=1, annot_kws={'size':12});
```

[+ Code](#)

Replacing 0s and 1s in Target column 'Churn'

```
] CC['Churn'] = CC['Churn'].replace( {'Churned': 1, 'Current': 0} )
```

```
<ipython-input-80-16c730ac43b1>:1: FutureWarning: Downcasting behavior in `replace` is deprecated
CC['Churn'] = CC['Churn'].replace( {'Churned': 1, 'Current': 0} )
```

```
] CC['Churn'].dtypes
```

```
dtype('int64')
```

Removal of unwanted variables - Dropping AccountID

```
] CC.drop(["AccountID","rev_growth_yoy", "coupon_used_for_payment"], axis=1, inplace=True)
```

```
] CC.shape
```

```
(11260, 16)
```

Machine Learning Algorithm

Logistic Regression Model - SKLearn

LR_model1 : Base model with outlier treated unscaled data

```
[ ] X = df.drop(columns=["Churn"])
    y = df["Churn"]
```

```
[ ] X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

```
[ ] LR_model1 = LogisticRegression(verbose=True,n_jobs=-1,random_state=123)
```

```
LR_model1.fit(X_train, y_train)
```

```
[Parallel(n_jobs=-1)]: Using backend LokyBackend with 2 concurrent workers.
```

```
LogisticRegression
LogisticRegression(n_jobs=-1, random_state=123, verbose=True)
```



```

def out_limit(col):
    Q1,Q3=col.quantile([0.25,0.75])
    IQR=Q3-Q1
    lower_range= Q1-(1.5 * IQR)
    upper_range= Q3+(1.5 * IQR)
    return lower_range, upper_range

for column in Outlier_treat_vars:
    lr,ur = out_limit(X[column])
    X[column] = np.where(X[column] > ur, ur, X[column])
    X[column] = np.where(X[column] < lr, lr, X[column])

plt.figure(figsize=(10,6))
sns.boxplot(data = X[Outlier_treat_vars]);

```

Outlier treatment

```
Outlier_treat_vars = ['Tenure', 'CC_Contacted_LY', 'rev_per_month', 'Day_Since_CC_connect', 'cashback']
```

```

Outlier_treat_vars = ['Tenure', 'CC_Contacted_LY', 'rev_per_month', 'Day_Since_CC_connect', 'cashback']
plt.figure(figsize=(10, 6))
sns.boxplot(data=X[Outlier_treat_vars])

```

```

LR_model1_ytrain_pred = LR_model1.predict(X_train)
LR_model1_ytest_pred = LR_model1.predict(X_test)

## Predicting probabilities on train dataset
LR_model1_trainprobs = LR_model1.predict_proba(X_train)
# # keep probabilities for the positive outcome only
LR_model1_trainprobs = LR_model1_trainprobs[:, 1]

## Predicting probabilities on test dataset
LR_model1_testprobs = LR_model1.predict_proba(X_test)
# # keep probabilities for the positive outcome only
LR_model1_testprobs = LR_model1_testprobs[:, 1]

# # calculate roc curve
train_auc = round(roc_auc_score(y_train, LR_model1_trainprobs),2)
test_auc = round(roc_auc_score(y_test, LR_model1_testprobs),2)
train_label_LR = 'LogReg Train AUC :' + str(train_auc)
test_label_LR = 'LogReg Test AUC :' + str(test_auc)

fpr_train, tpr_train, thresholds_train = roc_curve(y_train, LR_model1_trainprobs)
fpr_test_LR, tpr_test_LR, thresholds_test_LR = roc_curve(y_test, LR_model1_testprobs)

# # plot the roc curve for the model
plt.plot([0, 1], [0, 1], linestyle='--')
plt.plot(fpr_train, tpr_train, color='green',label=train_label_LR)
plt.plot(fpr_test_LR, tpr_test_LR, color='red', label=test_label_LR)
plt.legend(loc='best')
plt.title("AUC-ROC curve for train and test datasets")
plt.show();

```

```

from sklearn.metrics import roc_auc_score, roc_curve, classification_report, confusion_matrix, ConfusionMatrixDisplay

def model_eval(model,xtrue,ytrue,ypred,probs):

    plot_roc_curve(model,xtrue,ytrue,color='red') # Now plot_roc_curve should be recognized.
    plt.plot([0, 1], [0, 1], linestyle='--');

    print('Accuracy : ', model.score(xtrue, ytrue))

    # Instead of plot_confusion_matrix, use ConfusionMatrixDisplay
    ConfusionMatrixDisplay.from_estimator(model, xtrue, ytrue, colorbar=False).plot();

    print(classification_report(ytrue, ypred));

    # # calculate AUC
    auc = roc_auc_score(ytrue, probs)
    print('AUC: %.3f' % auc);

from sklearn.metrics import accuracy_score
log_acc = accuracy_score(y_train, LR_model1_ytrain_pred)

print("\nModel Performance:")
print(f"Logistic Regression Accuracy: {log_acc:.4f}")
print("\nClassification Reports:")
print("\nLogistic Regression:\n", classification_report(y_train, LR_model1_ytrain_pred))

cm = confusion_matrix(y_train, LR_model1_ytrain_pred)

# Display confusion matrix
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap='Blues') # You can change the colormap if needed
plt.title("Confusion Matrix - Training Data")
plt.show()

log_acc = accuracy_score(y_test, LR_model1_ytest_pred)

print("\nModel Performance:")
print(f"Logistic Regression Accuracy: {log_acc:.4f}")
print("\nClassification Reports:")
print("\nLogistic Regression:\n", classification_report(y_test, LR_model1_ytest_pred))

```

```

cm = confusion_matrix(y_test, LR_model1_ytest_pred)

# Display confusion matrix
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap='Blues') # You can change the colormap if needed
plt.title("Confusion Matrix - testing Data")
plt.show()

```

Support vector machine classifier

```

▶ from sklearn import svm

```

```

▶ def model_eval1(model,xtrue,ytrue,ypred):

    plot_roc_curve([model,xtrue,ytrue,color='red'])
    plt.plot([0, 1], [0, 1], linestyle='--');

    print('Accuracy : ', model.score(xtrue, ytrue))

    plot_confusion_matrix(model,xtrue,ytrue, colorbar=False);

    print(classification_report(ytrue, ypred));

```

```

[ ] from sklearn.preprocessing import StandardScaler

```

```

[ ] sc = StandardScaler()
    X_train = sc.fit_transform(X_train)
    X_test = sc.transform(X_test)

```

SVM_model1 : Default base model

```
[ ] ## Creating model and fitting train dataset
```

```
SVM_model1 = svm.SVC(random_state=1, probability=True)
SVM_model1.fit(X_train, y_train)
```



SVC

i ?

SVC(probability=True, random_state=1)

```
[ ] ### Predicting on train and test dataset
```

```
SVM_model1_ytrain_pred = SVM_model1.predict(X_train)
SVM_model1_ytest_pred = SVM_model1.predict(X_test)
```

```
[ ] ## Predicting probabilities on train dataset
```

```
SVM_model1_trainprobs = SVM_model1.predict_proba(X_train)
## keep probabilities for the positive outcome only
SVM_model1_trainprobs = SVM_model1_trainprobs[:, 1]
```

```
## Predicting probabilities on test dataset
```

```
SVM_model1_testprobs = SVM_model1.predict_proba(X_test)
## keep probabilities for the positive outcome only
SVM_model1_testprobs = SVM_model1_testprobs[:, 1]
```

```
## calculate roc curve
```

```
train_auc = round(roc_auc_score(y_train, SVM_model1_trainprobs),2)
```

```
test_auc = round(roc_auc_score(y_test, SVM_model1_testprobs),2)
```

```
train_label = 'Train AUC : ' + str(train_auc)
```

```
test_label = 'Test AUC : ' + str(test_auc)
```

```
fpr_train, tpr_train, thresholds_train = roc_curve(y_train, SVM_model1_trainprobs)
```

```
fpr_test, tpr_test, thresholds_test = roc_curve(y_test, SVM_model1_testprobs)
```

```
## plot the roc curve for the model
```

```
plt.plot([0, 1], [0, 1], linestyle='--')
```

```
plt.plot(fpr_train, tpr_train, color='green', label=train_label)
```

```
plt.plot(fpr_test, tpr_test, color='red', label=test_label)
```

```
plt.legend(loc='best')
```

```
plt.title("AUC-ROC curve for train and test datasets")
```

```
plt.show();
```



```

log_acc = accuracy_score(y_train, SVM_model1_ytrain_pred)

print("\nModel Performance:")
print(f"Logistic Regression Accuracy: {log_acc:.4f}")
print("\nClassification Reports:")
print("\nLogistic Regression:\n", classification_report(y_train, SVM_model1_ytrain_pred))

cm = confusion_matrix(y_train, SVM_model1_ytrain_pred)

# Display confusion matrix
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap='Blues') # You can change the colormap if needed
plt.title("Confusion Matrix - Training Data")
plt.show()

```

Random Forest

RF_model1: Base model without outliers and scaled data

```

from sklearn.ensemble import RandomForestClassifier

[ ] #Build a RandomForestClassifier and fit it on the training data
    RF_model1 = RandomForestClassifier(random_state=123, oob_score=True)
    RF_model1 = RF_model1.fit(X_train, y_train)

[ ] RF_model1.oob_score_

0.9761323268206039

[ ] ### Predicting on train and test dataset
    RF_model1_ytrain_pred = RF_model1.predict(X_train)
    RF_model1_ytest_pred = RF_model1.predict(X_test)

    ## Predicting probabilities on train dataset
    RF_model1_trainprobs = RF_model1.predict_proba(X_train)
    # # keep probabilities for the positive outcome only
    RF_model1_trainprobs = RF_model1_trainprobs[:, 1]

    ## Predicting probabilities on test dataset
    RF_model1_testprobs = RF_model1.predict_proba(X_test)
    # # keep probabilities for the positive outcome only
    RF_model1_testprobs = RF_model1_testprobs[:, 1]

```

```
# # calculate AUC and plot roc curve
train_auc = round(roc_auc_score(y_train, RF_model1_trainprobs),2)
test_auc = round(roc_auc_score(y_test, RF_model1_testprobs),2)
train_label = 'Train AUC :' + str(train_auc)
test_label = 'Test AUC :' + str(test_auc)

fpr_train, tpr_train, thresholds_train = roc_curve(y_train, RF_model1_trainprobs)
fpr_test, tpr_test, thresholds_test = roc_curve(y_test, RF_model1_testprobs)

# # plot the roc curve for the model
plt.plot([0, 1], [0, 1], linestyle='--')
plt.plot(fpr_train, tpr_train, color='green',label=train_label)
plt.plot(fpr_test, tpr_test, color='red', label=test_label)
plt.legend(loc='best')
plt.title("AUC-ROC curve for train and test datasets")
plt.show();
```

```
log_acc = accuracy_score(y_train, RF_model1_ytrain_pred)

print("\nModel Performance:")
print(f"Logistic Regression Accuracy: {log_acc:.4f}")
print("\nClassification Reports:")
print("\nLogistic Regression:\n", classification_report(y_train, RF_model1_ytrain_pred))

cm = confusion_matrix(y_train, RF_model1_ytrain_pred)

# Display confusion matrix
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap='Blues') # You can change the colormap if needed
plt.title("Confusion Matrix - Training Data")
plt.show()
```

Feature importance for the best SVM model

```
from sklearn.inspection import permutation_importance

[ ] imps = permutation_importance(SVM_model2, X_test, y_test)

[ ] imp_df = pd.DataFrame(imps.importances_mean, columns=['Permutation_importance'], index = X.columns)
## plotting graph
imp_df = imp_df.sort_values(by='Permutation_importance')
my_colors = ['red','green','blue','cyan','magenta','yellow']
imp_df.plot(kind='barh', color = my_colors, figsize=(8,8))
plt.title("Feature importance for SVM")
plt.show();
```

Feature importances for Random Forest Model

```
[ ] imp_df = pd.DataFrame(RF_model1.feature_importances_, index=X.columns, columns=['Feature Importance'])
    imp_df
```

```
## plotting graph
imp_df = imp_df.sort_values(by='Feature Importance')
my_colors = ['red','green','blue','black','orange','cyan','magenta','brown']
imp_df.plot(kind='barh', color = my_colors, figsize=(8,8))
plt.title("Feature importance using best model in Random Forest")
plt.show();
```

Comprehensive ROC plot for all models

```
## plot the roc curve for the model
plt.figure(figsize=(9,8))
plt.plot([0, 1], [0, 1], linestyle='--',label='Random classifier')
plt.plot(fpr_test_LR, tpr_test_LR, color='red', label=test_label_LR)
plt.plot(fpr_test_SVM, tpr_test_SVM, color='brown', label=test_label_SVM)
plt.legend(loc='best')
plt.xlabel("False positive rate")
plt.ylabel("True positive rate")
plt.title("ROC curve for test datasets across models")
plt.show();
```